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RESEARCH ARTICLE

WildfireKR: An LLM-Agent Decision Support Framework for Autonomous Wildfire Spread Simulation in South Korea

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ABSTRACT Wildfire management in South Korea requires rapid scenario exploration tools that enable responders to evaluate potential fire spread under varying weather and fuel conditions within operationally relevant timeframes. This paper presents WildfireKR, an open-source decision support system (DSS) that leverages large language model (LLM) agent orchestration to automate end-to-end wildfire spread simulation. The system chains ten functional modules, seven domain-specific and three supporting, via structured LLM tool-calling, enabling users to initiate multi-scenario analyses via natural language or structured YAML configurations. Korea-specific calibration comprising approximately 4,700 Monte Carlo simulations across four historical fires established a broadleaf spread probability of 0.70, reducing mean area error by 47% relative to the Mediterranean-calibrated baseline. Evaluation on three historical wildfires (Samcheok 2022, Goseong 2019, Uljin 2022) yielded area ratios of 0.65–3.58× relative to observed burned areas. The single underestimation (Goseong, 0.65×) is attributed to extreme wind-driven spotting beyond the simplified spotting model's capability, while the overestimations are consistent with the expected behavior of suppression-free simulations. Supplementary spatial assessment using Sentinel-2 differenced Normalized Burn Ratio (dNBR) derived burn scars with perimeter-based reference delineation (morphological closing) produced Recall values of 0.89–0.99 and IoU of approximately 0.24, indicating that the simulated fire envelope captured the majority of the actual burned area, a property relevant for decision support where false negatives carry disproportionate consequences. For comparison, pixel-based reference delineation yields lower IoU (0.13–0.16), demonstrating that much of the apparent spatial disagreement arises from interior unburned enclaves within fire boundaries rather than fundamental model failure. Systems evaluation confirmed high repeatability (burned-area CV = 3.3–7.1%), operational efficiency (2.7 min vs. 2–4 h manual workflow), and robustness (8/8 adversarial input tests detected). The framework will be released under the MIT license upon acceptance and comprises approximately 1,600 logical lines of Python.

INDEX TERMS Cellular automata, decision support system, fire spread modeling, large language model, South Korea, wildfire simulation.

I. INTRODUCTION

A. WILDFIRE DECISION SUPPORT CHALLENGES

Wildfire frequency and severity in South Korea have increased markedly over the past two decades, driven by warming spring temperatures, extended drought periods,

and the expansion of highly flammable *Pinus densiflora* plantations across eastern mountain corridors [1]. The 2022 Uljin–Samcheok wildfire complex—burning over 200 km² across nine days—demonstrated the inadequacy of reactive suppression alone and underscored the need for predictive decision support tools that can inform evacuation planning, resource pre-positioning, and tactical deployment.

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Effective wildfire decision support systems (DSS) must satisfy three operational requirements. First, rapid turnaround is essential, enabling responders to evaluate multiple “what-if” scenarios within minutes rather than hours. Second, accessibility to non-GIS-specialist users reduces the expertise barrier for scenario exploration. Third, transparent uncertainty communication ensures that probability-based outputs are interpreted as risk envelopes rather than deterministic predictions. Existing semi-empirical simulators such as FARSITE [2] and FlamMap [3], which employ Huygens’ wavelet principle with Rothermel’s empirical rate-of-spread equations rather than first-principles combustion physics [9], provide detailed fire behavior predictions but require extensive manual data preparation (fuel model parameterization, weather file formatting, landscape file construction, and projection alignment) that typically consumes 2–4 hours per scenario. This manual setup effort limits rapid multi-scenario exploration during emerging fire events. The core challenge is therefore not fire physics modeling, where decades of research have produced mature engines, but rather the *systems integration* required to make these engines operationally accessible under time pressure.

B. LLM AGENTS FOR SCIENTIFIC SIMULATION

The emergence of LLM-agent frameworks, where a language model orchestrates domain-specific computational tools through structured function calling, offers a pathway to bridge the accessibility–fidelity gap in environmental simulation. Recent work has demonstrated that LLM agents can autonomously operate scientific simulation software across multiple domains. EPANET-Agentica [4] orchestrates water distribution network simulation through a four-agent architecture comprising an orchestrator and three specialists (TaskExecutor, CodeRunner, DataAnalyzer), achieving fully autonomous natural-language-driven analysis on benchmark networks. Foam-Agent 2.0 [5] uses six specialized agents with Model Context Protocol (MCP)-based tool orchestration for CFD simulation. MetaOpenFOAM [6] automates CFD workflows through multi-agent RAG and collaborative code generation. These systems share a common pattern: the LLM translates user intent into structured tool calls, manages data flow between pipeline stages, and synthesizes results into interpretable outputs—all without requiring users to interact directly with the underlying simulation software.

The MCP specification [7] standardizes this tool-calling interface, enabling any compliant LLM to discover and invoke domain-specific computational modules. By exposing each pipeline stage as an MCP-compatible tool, the simulation framework becomes accessible to any MCP-enabled LLM client without framework-specific prompt engineering.

C. CONTRIBUTIONS

Although Table 1 surveys a growing body of LLM-agent simulation frameworks, wildfire spread modeling remains underexplored in this context. This paper presents WildfireKR,

an open-source decision support framework that addresses this gap with the following contributions:

- 1) An end-to-end automated DSS that chains ten functional modules via structured LLM tool-calling, enabling natural-language-driven wildfire simulation with approximately 50× speedup over manual workflows.
- 2) A Korea-specific calibration methodology comprising a three-stage hierarchical calibration (~4,700 simulations) that adapts the PROPAGATOR CA engine [8] from Mediterranean to Korean forest conditions.
- 3) Automated retrieval and processing of ERA5 reanalysis and Korea Meteorological Administration (KMA) station data with Nelson equilibrium moisture content (EMC) modeling for time-varying fuel moisture.
- 4) Multi-scenario analysis demonstrating the framework’s capability for parallel evaluation of hypothetical fire scenarios under varying weather conditions.
- 5) Supplementary spatial assessment using Sentinel-2 dNBR with both pixel-based and perimeter-based reference delineation, demonstrating that simulated fire envelopes capture the majority of observed burned areas.
- 6) A systems evaluation framework quantifying repeatability, operational efficiency, and guardrail robustness across multiple LLM backends.

The framework executes on consumer hardware (Apple M-series, 16 GB RAM) without GPU requirements.

The remainder of this paper is organized as follows. Section II reviews related work in CA fire modeling and LLM-agent frameworks. Section III describes the system architecture and module structure. Section IV details the methodology, including fuel classification, weather integration, and Korea-specific calibration. Section V presents experimental evaluation on historical fires, spatial assessment, and systems evaluation. Section VI discusses comparisons with existing frameworks, limitations, and future directions. Section VII concludes the paper.

II. RELATED WORK

A. CELLULAR AUTOMATA FIRE SPREAD MODELS

Cellular automata (CA) represent fire spread as stochastic local interactions on a discrete grid, offering computational simplicity compared to semi-empirical wave-propagation models such as FARSITE [2]. Sullivan [9] provides a comprehensive taxonomy of wildfire spread models; both CA and FARSITE fall within the “simulation and mathematical analogue” category—the former using calibrated spread probabilities on a discrete lattice, the latter employing Huygens’ wavelet principle with Rothermel’s semi-empirical rate-of-spread equations. Subsequent reviews by Papadopoulos and Pavlidou [24] and Andrianarivony and Akhloufi [30] have examined advances in CA and machine learning approaches to fire modeling since Sullivan’s taxonomy, noting increased integration of remote sensing data, ensemble

TABLE 1. Comparison of LLM-agent simulation frameworks.

Framework	Domain	Agents	Tool Interface	Evaluation	Year
EPANET-Agent [4]	Water networks	4	Function calling	Benchmark (100%)	2026
Foam-Agent 2.0 [5]	CFD	6	MCP	End-to-end test	2025
MetaOpenFOAM [6]	CFD	Multi	RAG + MetaGPT	Benchmark (85%)	2024
SGA [11]	Physics discovery	2	Code generation	Case studies	2024
WildfireKR	Wildfire	1	MCP	Systems + spatial	2026

methods, and deep learning-based spread prediction [26]. PROPAGATOR [8] is a probabilistic CA engine developed by the CIMA Research Foundation for Mediterranean fire risk assessment. It models fire spread through fuel-type-dependent base probabilities modified by slope, wind, and moisture factors, producing ensemble probability maps from multiple stochastic realizations. The engine has been operationally deployed in several Mediterranean countries [10] but has not previously been applied to East Asian forest ecosystems.

The distinction between CA and semi-empirical wave-propagation approaches (e.g., FARSITE) is relevant to the DSS context. CA models sacrifice detailed fire behavior predictions (flame length, rate of spread, crown fire transition) in exchange for rapid computation and intuitive probabilistic outputs. For decision support applications where the primary question is “which areas might be threatened?” rather than “what will the exact fire perimeter be?”, the ensemble probability map provides directly actionable information: areas with high probability (>0.5) warrant protective action regardless of the precise perimeter geometry.

B. LLM AGENT FRAMEWORKS FOR SIMULATION

Table 1 compares representative LLM-agent simulation frameworks as of early 2026.

WildfireKR adopts a single-agent architecture with MCP tool calling, similar to the approach used by Foam-Agent 2.0 [5] for CFD automation. This design simplifies error handling and state management compared to multi-agent systems such as MetaOpenFOAM [6] but limits the framework’s ability to self-validate or decompose complex reasoning tasks. The systems evaluation approach (repeatability, efficiency, guardrails) extends beyond the case-study-only evaluations common in the literature.

C. KOREAN WILDFIRE RESEARCH

Korean wildfire research has primarily focused on statistical risk mapping [12], post-fire damage assessment using remote sensing [13], and meteorological analysis of fire weather patterns [14]. Computational fire spread modeling for Korean landscapes remains limited, partly due to the absence of standardized Korean fuel models analogous to the Anderson 13 [15] or Scott and Burgan 40 [16] classifications used in the United States. The Korea Forest Service publishes aggregate fire statistics but does not provide georeferenced fire perimeter data in machine-readable formats, constraining spatial validation efforts. Recent work using Sentinel-2 dNBR (differenced Normalized Burn Ratio) for Korean

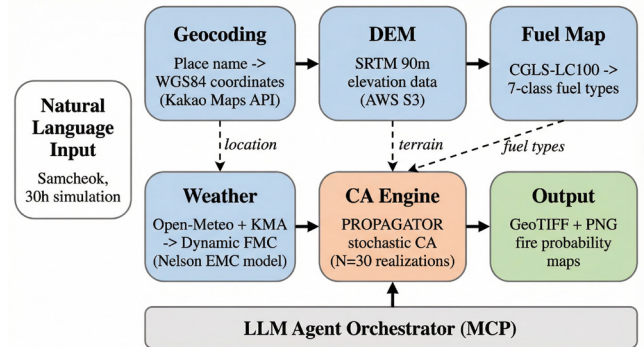


FIGURE 1. WildfireKR system architecture. Ten functional modules are organized in four layers: natural language interface, geospatial data acquisition, simulation engine, and output/visualization. All modules are exposed as MCP-compatible tools.

burn scar mapping [17] has begun to address this gap, enabling satellite-derived reference perimeters for model assessment. The dNBR index quantifies fire severity by comparing pre-fire and post-fire near-infrared and shortwave infrared reflectance, with thresholds established by Key and Benson [25] providing standardized severity classifications.

III. SYSTEM ARCHITECTURE

A. FRAMEWORK OVERVIEW

WildfireKR consists of ten functional modules organized in four layers (Fig. 1). The natural language interface layer parses user queries, resolves Korean place names via the Kakao Maps API geocoding service, and translates natural language descriptions to structured parameter configurations. The geospatial data acquisition layer downloads Shuttle Radar Topography Mission (SRTM) 1-arcsec digital elevation model (DEM) data, retrieves Copernicus Global Land Service (CGLS)-LC100 land cover for fuel classification, and fetches historical or real-time weather data from Open-Meteo (ERA5 reanalysis) or KMA Automated Surface Observing System (ASOS) stations. The simulation engine layer executes the PROPAGATOR CA engine with Korea-calibrated parameters, managing stochastic realizations and ensemble aggregation. The output and visualization layer generates georeferenced fire probability maps (GeoTIFF), visualizations (PNG with hillshade), structured metadata (JSON), and configuration records (YAML).

B. TOOL-BASED INTERACTION VIA MCP

Each module is exposed as an MCP-compatible tool that the LLM agent discovers and invokes through standardized function-calling interfaces. The tool schema defines

TABLE 2. Module structure of WildfireKR.

Module	LOC	Function	Dependencies
simulation.py	541	Pipeline orchestration	All modules
config.py	125	YAML/dict config schema	PyYAML
params.py	49	Korea-calibrated fuel params	PROPAGATOR
geocoder.py	68	Korean place-name geocoding	Kakao Maps API
data/srtm.py	84	SRTM DEM download	NumPy, requests
data/fuel.py	147	CGLS → fuel mapping	Rasterio, SciPy
data/weather.py	201	Weather + FMC modeling	Open-Meteo API
data/kma.py	92	KMA real-time weather	KMA API
output.py	223	GeoTIFF, PNG, JSON	Rasterio, Matplotlib
cli.py	58	CLI interface	Click

input parameters (location, weather conditions, simulation settings), output types (arrays, file paths, metadata), and validation constraints (parameter ranges, required fields). This architecture decouples the simulation logic from the LLM interface, allowing any MCP-enabled client to operate the framework without modification.

C. PIPELINE ALGORITHM

The end-to-end pipeline proceeds as follows:

When users provide imprecise specifications (e.g., “dry spring conditions”), the agent maps phrases to parameter ranges derived from calibration data and historical Korean weather records (e.g., “spring dry” → wind 8–12 km/h, FMC 10–14%). All resolved parameters are logged to the run record, enabling users to inspect, reproduce, or override them in subsequent YAML-based runs.

D. MODULE STRUCTURE

Table 2 summarizes the ten functional modules and their dependencies.

Line counts represent logical lines of code excluding blank lines and comments.

E. CONFIGURATION SYSTEM

The framework supports two interaction modes reflecting a two-tier trust model. In natural language mode, intended for rapid exploratory analysis, the LLM translates user intent to simulation parameters. This mode introduces parameter variability across sessions because the LLM may interpret identical natural language queries differently. In YAML mode, designed for reproducible and auditable analyses, all parameters are explicitly specified:

```
location:
  name: Samcheok
  lat: 37.27
  lon: 129.00
  radius_km: 15
simulation:
  cellsize: 90
  realizations: 30
  duration_h: 30
  bl_spread_prob: 0.70
  spotting: true
  fuel_source: cglis
scenarios:
  - name: Dynamic_Weather
```

```
weather_date: "2022-03-04"
weather_end_date: "2022-03-05"
```

For operational deployment, we recommend a human-on-the-loop protocol: the agent executes autonomously, but release of decision-relevant outputs (e.g., evacuation-support maps) requires analyst sign-off on key assumptions and run logs.

IV. METHODOLOGY

A. KOREAN FUEL CLASSIFICATION

No standardized Korean fuel classification analogous to the Anderson 13 or Scott and Burgan 40 models currently exists. We developed a mapping from CGLS-LC100 global land cover to PROPAGATOR’s 7-class fuel system (Table 3).

The mapping of mixed forest to conifers is a key adaptation that reflects the vertical fuel structure of Korean mountain forests. Even in mixed stands, the persistent needle-litter layer of *P. densiflora* provides the primary continuous fuel bed for surface fire propagation. During the dry spring fire season (March–May), when the vast majority of large Korean wildfires occur, deciduous oaks typically lack a fully developed canopy, allowing the volatile conifer litter to dominate the spread rate. Seo and Choung [18] demonstrated that *P. densiflora* litter ignites significantly faster and sustains combustion more reliably than broadleaf litter (e.g., *Quercus variabilis*), supporting this classification decision. This mapping is calibrated for the dominant spring fire season and may require adjustment for autumn fire events when deciduous canopy is fully developed. The conservative bias—overestimating spread in stands where broadleaf species truly dominate—is acceptable for DSS applications where overestimation is preferable to underestimation in evacuation planning.

B. WEATHER INTEGRATION AND DYNAMIC FMC

The framework integrates two weather data sources. The Open-Meteo Historical API provides hourly ERA5 reanalysis data with global coverage from 1940 at approximately 25 km resolution. The KMA API accesses 69 ASOS stations providing real-time observations, though mountain coverage is limited, with the nearest station potentially 20–50 km from fire zones.

Dynamic fuel moisture content (FMC) is computed through a three-component model. The equilibrium moisture

TABLE 3. CGLS-LC100 to PROPAGATOR fuel mapping for Korean landscapes.

CGLS Class (Code)	PROPAGATOR	Code	Rationale
Needle-leaf (111,113,121,123)	Conifers	5	Direct match
Deciduous broad-leaf (114,124)	Broadleaves	1	Direct match
Mixed/Unknown (115,116,125,126)	Conifers	5	Korean mixed $\approx$ conifer-dominant
Shrubland (20)	Shrubs	2	Direct match
Herbaceous (30)	Grassland	4	Direct match
Cultivated (40)	Agro-forestry	6	Moderate susceptibility
Bare/Sparse (60)	Grassland	4	Sparse vegetation combustible
Urban/Snow/Water (50,70,80)	Non-vegetated	3	Fire barriers

Algorithm 1 WildfireKR End-to-End Pipeline**Require:** Natural language query Q or YAML configuration**Ensure:** Fire probability map $\bar{P}(x, y) \in [0, 1]$

```

1:  $(\varphi, \lambda, r) \leftarrow \text{Geocode}(Q)$   $\triangleright$  Resolve place name
2:  $Z \leftarrow \text{Download\_DEM}(\varphi, \lambda, r)$   $\triangleright$  SRTM 1-arcsec
3: Resample  $Z$  to cell size  $\Delta x$  (default 90 m)
4:  $F \leftarrow \text{Classify\_Fuel}(\varphi, \lambda, r)$   $\triangleright$  CGLS  $\rightarrow$  7-class Korea mapping
5: Resample  $F$  to match  $Z$  grid
6:  $(i_0, j_0) \leftarrow \text{Find\_Ignition}(F, Z)$   $\triangleright$  Vegetated cell nearest center
7: if historical weather mode then
8:    $W \leftarrow \text{Fetch\_Weather}(\varphi, \lambda, t_{\text{start}}, t_{\text{end}})$   $\triangleright$  Hourly weather
9:    $M \leftarrow \text{Compute\_FMC}(W)$   $\triangleright$  Dynamic FMC (Nelson EMC)
10:   $BC \leftarrow \text{Make\_Boundary\_Conditions}(W, M)$   $\triangleright$  Time-varying BCs
11: else
12:   $BC \leftarrow \text{Make\_Static\_BC}(\Theta)$   $\triangleright$  User-specified
13: end if
14:  $P_{\text{fuel}} \leftarrow \text{Load\_Calibrated\_Params}()$   $\triangleright$  Korea-tuned  $p_{\text{BL}} = 0.70$ 
15: for  $k = 1$  to  $N$  do  $\triangleright N = 30$  realizations
16:    $B_k \leftarrow \text{Propagator\_CA}(Z, F, BC, P_{\text{fuel}}, i_0, j_0)$ 
17: end for
18:  $\bar{P} \leftarrow (1/N) \sum B_k$   $\triangleright$  Ensemble mean
19: Save_Outputs( $\bar{P}, Z$ , formats={GeoTIFF, PNG, JSON})
20: return  $\bar{P}$ 

```

content (EMC) follows Nelson [19] and Simard [20], with relative humidity expressed as a fraction (0–1) and EMC returned as a fraction of dry weight:

$$\text{EMC}(RH) = \begin{cases} 0.03 + 0.20 \cdot RH & RH < 0.10 \\ 0.05 + 0.16 \cdot RH & 0.10 \leq RH < 0.50 \\ 0.20 + 0.40(RH - 0.50) & RH \geq 0.50 \end{cases} \quad (1)$$

The time-lag response follows Van Wagner [21]:

$$M(t) = \text{EMC}(t) + [M(t-1) - \text{EMC}(t)] \cdot \exp(-\Delta t / \tau) \quad (2)$$

where $\tau = 10$ h (10-hour fuels) and $\Delta t = 1$ h. The precipitation wetting component, also from Van Wagner [21], is:

$$\Delta M = 10.625 \cdot r_{\text{eff}} \cdot \exp\left(\frac{-25}{101 - M}\right) \cdot \left(1 - \exp\left(\frac{-6.93}{r_{\text{eff}}}\right)\right) \quad (3)$$

where $r_{\text{eff}} = \max(P - 0.5, 0)$ mm. A 7-day conditioning period precedes the simulation to equilibrate FMC from an initial 15% baseline, ensuring the moisture state at ignition reflects cumulative recent meteorological history rather than an arbitrary initial condition. The three-piece EMC formulation was assembled from Nelson [19] (low-humidity regime) and Simard [20] (moderate and high-humidity regimes).

C. KOREA-SPECIFIC CALIBRATION

PROPAGATOR's default (LEGACY) parameters were calibrated for Mediterranean vegetation. Korean forests differ substantially in fuel bed density, terrain steepness, and the high flammability of *P. densiflora* litter.

The calibration was performed using a three-stage hierarchical approach comprising approximately 4,700 total simulations. The objective function minimized the mean absolute log-ratio of simulated to observed burned area:

$$\text{Obj}(p) = \frac{1}{K} \sum_{k=1}^K \left| \log \frac{A_{\text{sim},k}(p)}{A_{\text{obs},k}} \right| \quad (4)$$

where $K = 4$ cases (Samcheok 2022, Goseong 2019, Uljin 2022, Andong 2020), and $A_{\text{sim},k}$ is the ensemble mean burned area (cells with $\bar{P} > 0.5$) from $N = 30$ realizations per case.

Stage 1 tested p_{BL} at 10 values across 4 historical cases with $N = 30$ realizations each ($10 \times 4 \times 30 = 1,200$ simulations). Three of the four calibration cases (Samcheok, Goseong, Uljin) also serve as demonstration cases in Section V; the fourth (Andong 2020, 1.94 km²) was used only for calibration. The objective function is flat from $p_{\text{BL}} = 0.65$ to 0.70 (both $\text{Obj} \approx 0.49$) and decreases further at $p_{\text{BL}} = 0.75$ ($\text{Obj} = 0.34$), confirming that the parameter is robust across the studied regimes. We adopt $p_{\text{BL}} = 0.70$ as a conservative choice within the stable plateau. Stages 2 and 3 showed negligible sensitivity (less than 5% change in Obj), so LEGACY values were retained.

TABLE 4. Three-stage hierarchical calibration results.

Stage	Parameter	Range	Sims	Result
1	BL spread prob (p_{BL})	0.30–0.75, step 0.05	1,200	0.70
2	Base spread velocity	$\pm 50\%$ from LEGACY	$\sim 2,400$	Unchanged
3	FMC scale factor	0.5–1.5	$\sim 1,080$	1.0 (unchanged)

TABLE 5. Calibration results: LEGACY vs. Korea-tuned parameters.

Parameter	LEGACY	Korea-Tuned
p_{BL}	0.30	0.70
BL velocity (m/h)	140	140 (unchanged)
Conifer velocity (m/h)	200	200 (unchanged)
FMC scale	1.0	1.0 (unchanged)
Mean area error (Obj)	0.94	0.50 (–47%)

TABLE 6. Leave-one-out cross-validation results.

Held-out case	Actual (km ²)	Opt. p_{BL}	Sim. (km ²)	$ \log R $
Samcheok 2022	14.70	0.75	10.94	0.30
Goseong 2019	5.29	0.75	3.91	0.30
Uljin 2022	23.60	0.75	26.51	0.12
Andong 2020	1.94	0.75	1.00	0.66
Mean LOOCV		0.75		0.34

R = simulated/observed area ratio. All four folds select $p_{BL} = 0.75$; mean LOOCV error (0.34) equals the in-sample error. Adopted $p_{BL} = 0.70$ lies within the flat objective plateau (Section IV-C).

The $2.33\times$ increase in p_{BL} reflects Korean forests' higher fire susceptibility compared to Mediterranean maquis, consistent with the measured high flammability of *P. densiflora* needle litter [18].

Because the population of georeferenced Korean fire events is small, all four cases were used for calibration. The case study results in Section V-B therefore quantify calibration fit (internal consistency). To assess potential overfitting, we performed leave-one-out cross-validation (LOOCV) across the four calibration cases: in each fold, one case was held out, the optimal p_{BL} was determined from the remaining three cases, and the held-out case was evaluated at that optimum (Table 6).

All four folds independently selected $p_{BL} = 0.75$ as optimal, and the mean LOOCV error (0.34) was identical to the in-sample error at $p_{BL} = 0.75$, indicating no observable degradation from overfitting. This stability reflects the homogeneity of the Korean fire regime—dominated by spring *P. densiflora* fires on similar terrain—where a single parameter captures the dominant fuel-type effect across diverse fire sizes and durations. The LOOCV-optimal value (0.75) is slightly higher than the adopted $p_{BL} = 0.70$, providing additional conservative margin. We note that the four calibration events span substantially different scales (1.94–23.6 km²) and durations (12–213 h), and the primary contribution is the agentic framework and its calibratability rather than a validated predictive model. We highlight the need for a systematic, machine-readable Korean fire perimeter database to enable independent validation.

To further assess generalizability beyond the four calibration cases, we conducted an extended area-ratio sweep

across 10 Korean wildfire events (4 tuning + 6 holdout) at 10 p_{BL} values (0.30–0.75), yielding 3,000 total simulations (10 cases $\times$ 10 configurations $\times$ 30 realizations). All simulations used a standardized domain (15 km radius, auto-placed ignition) to ensure comparable conditions. Of the 10 cases, seven produced physically meaningful fire spread at $p_{BL} = 0.70$, with area ratios spanning $0.50\text{--}1.31\times$ observed (mean $|\log R| = 0.38$ for tuning, 0.21 for the three successful holdout cases). Three holdout cases (Gangneung 2022, Gangneung 2019, Andong-Yecheon 2021) produced near-zero ratios ($< 0.02\times$), attributable to ignition placement errors or multi-point fire origins not captured by the single auto-placed ignition. The mean absolute log-ratio error across all 10 cases was minimized at $p_{BL} = 0.70\text{--}0.75$ (Obj = 0.51–0.51), consistent with the four-case calibration result. Crucially, reducing p_{BL} below 0.55 caused catastrophic under-prediction for the majority of cases (mean ratio < 0.30), confirming that the adopted $p_{BL} = 0.70$ lies at the lower boundary of the operational plateau and cannot be reduced to decrease spatial overestimation without severely degrading area calibration.

D. CA SIMULATION ENGINE

Fire spread probability from cell i to adjacent cell j is computed as:

$$P_{ij} = p_0(f_i, f_j) \cdot \varphi_s(\theta_{ij}) \cdot \varphi_w(\vec{W}, \vec{d}_{ij}) \cdot \varphi_m(M) \quad (5)$$

where p_0 is the fuel-dependent base probability from the PROPAGATOR LEGACY transition matrix (e.g., broadleaf-to-broadleaf = 0.30, conifer-to-conifer = 0.35 in the Mediterranean default; the Korea calibration scales only broadleaf-related transitions to $p_{BL} = 0.70$ while conifer-to-conifer remains at 0.35); φ_s is the slope factor, which varies linearly from $0.5\times$ at 30° downhill to $3\times$ at 30° uphill, with flat terrain yielding $1.0\times$; φ_w is the wind factor, reaching up to $2\times$ for cells aligned with the wind direction; and φ_m is the moisture factor, which approaches 0 at $FMC \geq 30\%$.

Each time step (1 hour), every burning cell evaluates its eight neighbors (Moore neighborhood). For each unburned vegetated neighbor j , a uniform random number $u \sim U(0, 1)$ is drawn; if $u < P_{ij}$, cell j ignites. Over N realizations (default 30), the ensemble mean probability map $\bar{P}(x, y) = (1/N) \sum B_k(x, y) \in [0, 1]$ is computed, where $B_k(x, y) = 1$ if cell (x, y) burned in realization k .

The $\bar{P} > 0.5$ threshold implements a majority-rule criterion: a cell is classified as burned if it ignited in more than half of all realizations. This threshold provides a balance between capturing the high-probability fire core and excluding stochastic outlier cells. A simplified stochastic

ember spotting mechanism allows burning cells in high-wind conditions to ignite a random cell 2–5 cells downwind with a default probability of 0.5%. This captures long-range fire transport observed in Korean mountain fires during strong foehn events, though the default spotting parameters were adopted from PROPAGATOR’s Mediterranean configuration and may underestimate Korean foehn-driven spotting distances.

E. BURN SCAR REFERENCE DELINEATION

Spatial assessment of fire spread simulations requires a reference burned area derived from satellite imagery. The standard pixel-level approach classifies individual cells exceeding a dNBR threshold (e.g., $\text{dNBR} > 0.27$) as burned, producing scattered pixel clusters rather than a continuous fire boundary. However, real fire perimeters are contiguous—even when interior pockets remain unburned due to topographic barriers, fuel breaks, or suppression effects. Comparing a contiguous simulated fire envelope against scattered reference pixels inflates the apparent false-negative rate, because unburned interior areas within the actual fire perimeter are counted as simulation misses.

To address this representational mismatch, we implement a perimeter-based reference delineation that approximates the manually digitized burn boundaries used by the U.S. Monitoring Trends in Burn Severity (MTBS) program [31]. The delineation proceeds in three steps:

- 1) **Noise filtering:** Connected components with fewer than 10 pixels are removed to suppress isolated dNBR artifacts (e.g., cloud shadows, terrain-induced reflectance anomalies).
- 2) **Morphological closing:** A binary closing operation (dilation followed by erosion) with a disk-shaped structuring element of radius r pixels bridges narrow gaps between dNBR clusters. This connects nearby burned patches while preserving the overall boundary shape.
- 3) **Hole filling:** Binary hole filling (`scipy.ndimage.binary_fill_holes`) produces a continuous reference perimeter by filling interior unburned enclaves within the closed boundary.

The closing radius r is the single free parameter controlling the degree of gap bridging. We test $r \in \{1, 3, 5, 7\}$ pixels (corresponding to 30–210 m for Goseong at 30 m resolution, and 90–630 m for Uljin at 90 m resolution). The sensitivity of IoU to r is reported in Table 10 and Fig. 4a. For both cases, IoU increases monotonically with r , reflecting progressive inclusion of interior unburned areas within the reference perimeter. At $r = 7$, the reference area approximately doubles relative to the pixel-only baseline (Goseong: $8.0 \rightarrow 15.0 \text{ km}^2$; Uljin: $66.4 \rightarrow 94.5 \text{ km}^2$), consistent with the observation that large fire perimeters routinely contain 30–50% interior unburned patches. Crucially, Recall is preserved or slightly improved because the perimeter-filled reference encompasses the original dNBR pixels; the IoU

gains derive entirely from reclassifying interior false negatives as true negatives (areas correctly outside the reference) rather than from artificially inflating the overlap.

V. EXPERIMENTS AND EVALUATION

Because the primary contribution is an agentic DSS framework rather than a novel fire physics model, the evaluation is organized in two complementary parts. First, **simulation assessment** (Sections V-A through V-D) characterizes fire behavior output—area ratios quantify calibration fit (not out-of-sample predictive skill) and spatial metrics assess coverage of actual burn scars. Second, **systems evaluation** (Section V-E) measures the properties unique to an automated DSS: repeatability, operational efficiency, and guardrail robustness. Readers should interpret the simulation results as characterizing what the CA engine produces when embedded in the automated pipeline, rather than as claims of predictive accuracy comparable to semi-empirical models such as FARSITE.

A. EXPERIMENTAL SETUP

Three calibration cases (Samcheok 2022, Goseong 2019, Uljin 2022) plus one multi-scenario site (Palgongsan) were simulated using 90 m cells, $N = 30$ realizations, CGLS fuel classification, and the Korea-calibrated $p_{BL} = 0.70$. A fourth calibration fire (Andong 2020, 1.94 km^2) is not reported as a separate case study due to its small scale. To assess temporal generalizability, a holdout case—the Uiseong–Andong–Yeongyang 2025 mega-fire (hereafter “Uiseong 2025”), one of the largest Korean wildfires on record—was simulated using identical pipeline settings and parameters calibrated exclusively on 2019–2022 events. All simulations were executed on a consumer laptop (Apple M-series, 16 GB RAM) without GPU, using Numba JIT compilation for CA engine acceleration.

B. HISTORICAL CASE STUDIES

Three historical wildfires spanning a range of sizes (5–185 km^2), durations (12–213 h), and landscape conditions were used as calibration cases, supplemented by one holdout case for temporal generalizability assessment (Table 7).

The Samcheok 2022 fire burned approximately 14.7 km^2 over 30 hours beginning in March 2022. The simulation domain was centered at 37.27°N , 129.00°E with $r = 15 \text{ km}$, producing a 324×203 grid. The fuel composition was 66.4% broadleaf and 30.2% conifer, with FMC at ignition of 13.8%. The simulation produced 52.60 km^2 ($3.58 \times$ observed). This overestimation is attributed to three structural factors: suppression omission (the actual response included 28 helicopters and thousands of ground personnel), the 90 m cell size that cannot resolve narrow fire barriers, and CGLS fuel classification that overestimates continuous fuel coverage.

The Goseong 2019 fire burned approximately 5.29 km^2 over 12 hours in April 2019. The simulation domain at 38.20°N , 128.60°E produced a 324×413 grid in which 54.5%

of cells were classified as non-vegetated (coastal region). The simulation produced 3.44 km^2 ($0.65 \times$ observed). The high non-vegetated fraction acted as a natural fire barrier, constraining simulated spread. The underestimation, despite the absence of modeled suppression, is explained by extreme winds (sustained speeds exceeding 20 m/s with gusts above 30 m/s) that enabled spot fires over non-vegetated patches—a long-range transport mechanism only partially captured by the simplified spotting model. During this event, the Korean Meteorological Administration issued a strong wind advisory for the Goseong region, underscoring the anomalous wind conditions that drove fire behavior beyond the model's spotting parameterization.

The Uljin 2022 fire, one of the largest in modern Korean history, burned approximately 184.63 km^2 (Uljin county portion) over 213 hours (approximately 9 days) beginning March 4, 2022. The simulation domain was centered at 36.99°N , 129.40°E with $r = 35 \text{ km}$. The simulation produced 393.35 km^2 ($2.13 \times$ observed). The lower ratio compared to Samcheok ($3.58 \times$) likely reflects that the fire's extreme scale reduced the relative effectiveness of suppression efforts, narrowing the gap between suppressed reality and unsuppressed simulation.

1) HOLDOUT CASE: UISEONG 2025

The Uiseong–Andong–Yeongyang mega-fire of March 2025 burned approximately 993 km^2 across five municipalities over 213 hours, making it one of the largest wildfires on record in South Korea. The event involved at least 10 distinct ignition points with staggered onset times spanning 11–177 hours after the initial ignition in Uiseong-gun, driven by sustained gust winds exceeding 20 m/s during the early propagation phase. The simulation was configured with 10 staggered ignition points, gust-adjusted wind fields, and a domain radius of 50 km centered at 36.35°N , 128.97°E , using the same $p_{\text{BL}} = 0.70$ and 90 m cell size calibrated exclusively on 2019–2022 events. The simulation produced 1172.6 km^2 ($1.18 \times$ the official observed area). This near-unity ratio—substantially lower than the calibration-case overestimation range (0.65 – $3.58 \times$)—is consistent with the expectation that suppression-free simulations converge toward observed area as fire size increases, because organized suppression becomes proportionally less effective against mega-fires. Spatial assessment using Sentinel-2 dNBR (moderate-to-high severity threshold 0.27) yielded $\text{IoU} = 0.231$, $\text{Recall} = 0.439$, and $\text{Precision} = 0.327$ (Table 9). The lower Recall compared to the calibration cases (Goseong 0.97 – 0.99 , Uljin 0.89 – 0.91) is attributed to the multi-front nature of the event: the staggered ignition configuration, while informed by official fire reports, may not capture secondary ignitions from spot fires and re-ignitions that contributed to the actual burned perimeter.

For the three calibration cases, area ratios span a factor of approximately six (0.65 – $3.58 \times$; ratio of extremes = 5.5), which is wide but consistent with the structural bias drivers that apply across all cases. Suppression omission introduces

positive bias, simplified spotting introduces negative bias, ignition-location uncertainty contributes bidirectional bias, and coarse fuel and DEM resolution introduces positive bias. The observed ratios reflect the net balance of these competing drivers. The holdout case (Uiseong 2025, $1.18 \times$) exhibits the lowest overestimation, supporting the hypothesis that suppression-free CA simulations converge toward observed area for mega-fires where organized suppression is overwhelmed. Across all four cases, the opposing error directions are consistent with the identified structural mechanisms.

C. SPATIAL ASSESSMENT WITH SENTINEL-2 dNBR

To complement the area-level comparison, spatial assessment was conducted using Sentinel-2 dNBR [25] for the two calibration cases with available cloud-free imagery (Goseong 2019 and Uljin 2022) and for the holdout case (Uiseong 2025).

Burn scar reference perimeters were derived using the moderate-to-high severity threshold of $\text{dNBR} > 0.27$, following the severity classification of Key and Benson [25]. Simulated burn areas were defined by the ensemble probability threshold $\bar{P} > 0.5$ (majority rule). For the coastal Goseong domain, a $\text{DEM} > 0$ land mask was applied to exclude ocean pixels exhibiting spurious dNBR values. Fig. 2 visualizes the resulting spatial overlay for both cases, showing the simulated fire probability field against the dNBR-derived burn scar boundary on a hillshaded DEM background.

To ensure consistency with the area calibration, the spatial assessment was first conducted using the same calibrated parameter $p_{\text{BL}} = 0.70$ (Table 9, “Calibrated” rows). At this setting, the simulated fires substantially overestimate the perimeter-based dNBR burn scar area (approximately $7 \times$ for Goseong, $5 \times$ for Uljin) but achieve high Recall (0.99 and 0.91 , respectively), confirming that the calibrated model captures the spatial footprint of actual fires. The Goseong case was simulated for 21 hours (the full event duration, compared to the 12-hour primary spread phase used in Section V-B) to enable comparison with post-fire satellite imagery that captures the complete burn scar; the Uljin simulation covers the full 213-hour event. The substantial difference between Goseong's 12-hour area ratio ($0.65 \times$, Table 7) and the 21-hour spatial assessment ($\sim 7 \times$ at $p_{\text{BL}} = 0.70$) reflects primarily the extended duration—which allows the fire to spread well beyond the 12-hour primary perimeter—and underscores the high sensitivity of CA models to simulation duration.

For operational DSS deployment where spatial coverage is paramount, a coverage-optimized setting of $p_{\text{BL}} = 0.50$ was identified through a parameter sweep (Table 8) as the lowest value achieving $\text{Recall} \approx 0.97$ for the Goseong case (Table 9, “Coverage-opt.” rows). Because all fuel-type spread probabilities scale proportionally to p_{BL} (via the ratio $p_{\text{BL}}/0.30$), this lower value reduces all spread probabilities and produces a smaller simulated fire while maintaining high spatial coverage. At $p_{\text{BL}} = 0.50$, Goseong overestimation decreases to approximately $4 \times$ while Recall remains 0.97 ;

TABLE 7. Historical case study results ($p_{BL} = 0.70$, $N = 30$, 90 m cells).

Case	Role	Observed (km ²)	Simulated (km ²)	Ratio	Duration (h)
Samcheok 2022	Calibration	14.7	52.60	3.58×	30
Goseong 2019	Calibration	5.29	3.44	0.65×	12
Uljin 2022	Calibration	184.63	393.35	2.13×	213
Uiseong 2025†	Holdout	~993	1172.6	1.18×	213
Oak Fire 2022‡	Generalizability	77.9	173.4	2.23×	72

†Holdout case: parameters were calibrated exclusively on 2019–2022 events. Simulation used 10 staggered ignition points with gust-adjusted wind fields over 213 hours. Observed area from Korea Forest Service official estimate. ‡Geographic generalizability case: Mariposa County, California, USA (37.55°N, 119.92°W). Sierran mixed conifer forest. $p_{BL} = 0.30$ (Mediterranean default); ERA5 weather via Open-Meteo with America/Los_Angeles timezone. Observed area from CAL FIRE FRAP. Code changes required: timezone parameterization only (~5 LOC). The 2.23× overestimation is consistent with Korean cases and reflects suppression omission rather than framework failure.

Uljin achieves the best IoU (0.239) with Recall = 0.89. Crucially, both the calibrated ($p_{BL} = 0.70$) and coverage-optimized ($p_{BL} = 0.50$) settings are reported for transparency (Table 9); the choice between them reflects the user's risk tolerance for the over-warning–under-warning trade-off rather than a model inconsistency.

A systematic parameter sweep (Table 8, Fig. 3a,b) explored $p_{BL} = 0.30$ –0.70 for Goseong and $p_{BL} = 0.50$ –0.80 for Uljin. For Goseong, Recall reaches approximately 0.97 at $p_{BL} = 0.50$ and 0.99 for $p_{BL} \geq 0.60$, while IoU peaks at $p_{BL} = 0.40$ (IoU = 0.303); $p_{BL} = 0.50$ was selected to balance spatial coverage against area overestimation. For Uljin, Recall peaks at $p_{BL} = 0.70$ (0.912) but at the cost of substantial overestimation (4.7× the perimeter-based reference area); $p_{BL} = 0.50$ provides the best IoU (0.239) with Recall = 0.89. The Uljin sweep exhibits non-monotonic area and Recall behavior (e.g., area decreasing from 342.1 km² at $p_{BL} = 0.50$ to 250.9 km² at $p_{BL} = 0.60$ before increasing to 442.3 km² at $p_{BL} = 0.70$), reflecting the interaction between stochastic ensemble variability, terrain-dependent spread pathways, and the majority-rule ($\bar{P} > 0.5$) threshold over 213-hour simulations where the fire has sufficient time to interact with domain-scale topographic features.

Table 10 compares the pixel-based and perimeter-based reference delineation for the coverage-optimized $p_{BL} = 0.50$ setting. The perimeter approach (closing radius $r = 7$) improves IoU by 88% for Goseong (0.127 → 0.238) and 54% for Uljin (0.155 → 0.239), while Recall is maintained or slightly improved (Goseong: 0.960 → 0.966; Uljin: 0.825 → 0.892). This improvement arises because the perimeter reference correctly excludes interior unburned patches from the false-negative count, providing a fairer comparison for a contiguous simulated fire envelope. The perimeter-based IoU of approximately 0.24 for both cases represents the approximate upper bound achievable for a suppression-free simulation at approximately 4× area overestimation relative to the perimeter reference (theoretical maximum IoU $\approx r/\alpha \approx 0.97/4 \approx 0.24$ for perimeter reference vs. $0.97/7 \approx 0.14$ for pixel reference, where α is the overestimation ratio relative to the respective reference area).

Fig. 4 shows the sensitivity of IoU to the closing radius r and to the fire probability threshold under both reference

delineation methods. The monotonic increase in IoU with r (Fig. 4a) demonstrates that the improvement is robust and not an artifact of a specific radius choice. Under the probability threshold sweep (Fig. 4b), the perimeter-based IoU consistently exceeds the pixel-based IoU across all thresholds, with the gap largest at moderate thresholds ($\tau = 0.4$ –0.6) where the simulated area best balances coverage against overestimation. The optimal probability threshold shifts slightly under perimeter reference (Goseong: $\tau^* = 0.60$ for perimeter vs. 0.70 for pixel), but the practical difference is modest (perimeter IoU = 0.249 at $\tau = 0.60$ vs. 0.242 at $\tau = 0.70$).

Uncertainty bounds ($\pm$) represent 95% confidence intervals derived from parametric bootstrap resampling (1,000 iterations) of the $N = 30$ Monte Carlo realizations, treating each cell's fire probability as the success parameter of a Binomial(30, $\hat{p}$) distribution. These conditional stochastic confidence intervals capture Monte Carlo sampling variability but do not account for spatial autocorrelation among neighboring cells or uncertainty in the dNBR burn scar delineation threshold; the wider sensitivity to dNBR threshold choice is quantified separately in Table 19. The narrow confidence intervals confirm that $N = 30$ realizations yield stable spatial metrics for the majority-rule threshold.

The Recall values of 0.89–0.99 indicate that the simulated fire envelope captured 89–99% of the perimeter-delineated burned area as identified by satellite imagery. For DSS applications, Recall is the primary spatial metric because false negatives (unwarned areas that actually burn) carry substantially higher consequences than false positives (warned areas that do not burn) in evacuation and resource allocation decisions. Calkin et al. [22] have documented the severe consequences of under-warning in wildfire events, and the general principle that costs of missed warnings exceed costs of over-warning is well established in the natural hazard literature [23]. The IoU values (0.15–0.24) are moderate in absolute terms, partially reflecting the substantial area overestimation inherent in suppression-free simulations and the choice of a moderate-to-high severity dNBR threshold (Table 19). This trade-off is deliberate: by prioritizing Recall (spatial coverage) over Precision (spatial accuracy), the

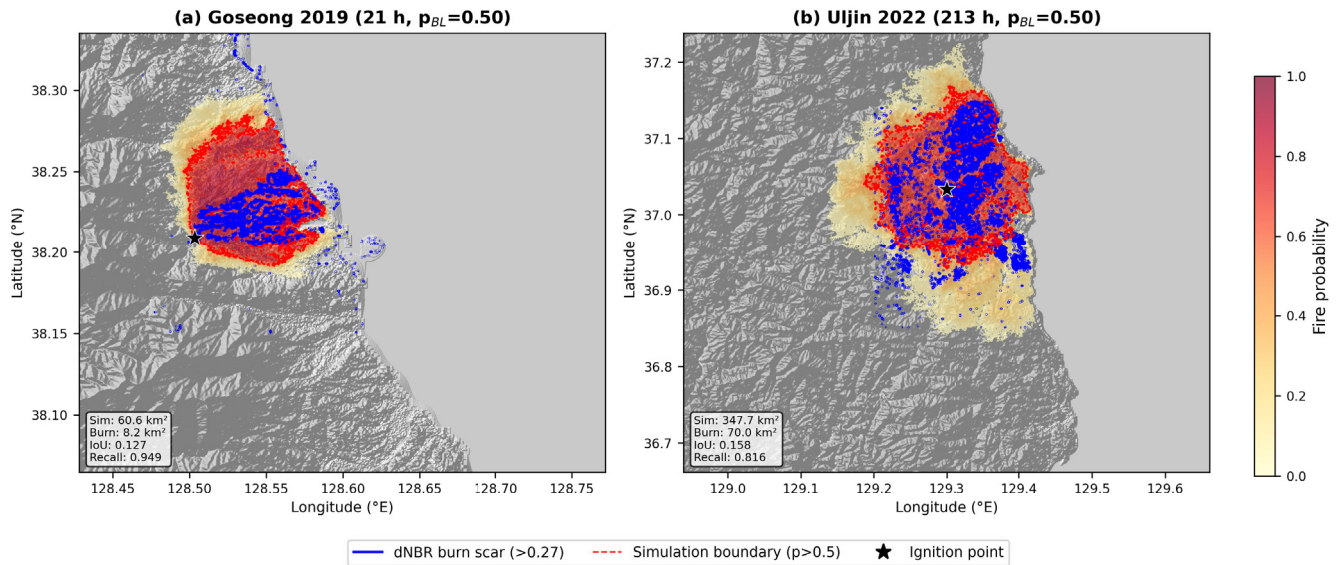


FIGURE 2. Spatial assessment overlay for (a) Goseong 2019 and (b) Uljin 2022.

TABLE 8. p_{BL} sensitivity sweep for spatial assessment cases ($N = 30$ realizations, perimeter-based dNBR > 0.27 reference, closing radius $r = 7$).

Case	p_{BL}	Sim. Area (km ²)	Ratio [†]	IoU	Precision	Recall
Goseong 21 h	0.30	0.03	—	—	—	—
Goseong 21 h	0.40	27.54	1.83×	0.303	0.360	0.659
Goseong 21 h	0.50	60.56	4.03×	0.238	0.240	0.966
Goseong 21 h	0.60	80.83	5.37×	0.183	0.183	0.985
Goseong 21 h	0.70	100.32	6.67×	0.149	0.149	0.993
Uljin 213 h	0.50	342.07	3.62×	0.239	0.246	0.892
Uljin 213 h	0.60	250.89	2.65×	0.173	0.203	0.539
Uljin 213 h	0.70	442.26	4.68×	0.191	0.195	0.912
Uljin 213 h	0.80	404.64	4.28×	0.166	0.176	0.753

[†]Ratio = Sim. Area / perimeter-based dNBR reference area (Goseong: 15.04 km²; Uljin: 94.53 km²). All metrics use perimeter-based reference delineation (morphological closing $r = 7$, Section IV-E). The selected $p_{BL} = 0.50$ setting (used in Table 9) is marked by asterisk-free entries at 0.50 for each case. For Goseong, $p_{BL} = 0.30$ falls below the percolation threshold and produces negligible fire spread. IoU peaks at $p_{BL} = 0.40$ for Goseong (0.303) and at $p_{BL} = 0.50$ for Uljin (0.239). Recall reaches approximately 0.97 for Goseong at $p_{BL} = 0.50$ (0.966). The Uljin sweep exhibits non-monotonic behavior (area decreasing from $p_{BL} = 0.50$ to 0.60 then increasing at 0.70), reflecting stochastic ensemble variability in the 213-hour simulation; area saturation occurs at $p_{BL} \geq 0.70$, where the simulated area exceeds $4.5\times$ the perimeter-based reference area.

system correctly identifies most threatened areas at the cost of also flagging regions where suppression prevented actual burning. For applications where missing a threatened area is far more costly than over-warning, this asymmetric error profile is operationally preferable to a balanced model that might underestimate the threatened zone.

The dual-parameter reporting (Table 9) reflects the inherent tension in DSS design between calibration fidelity and operational coverage. The calibrated $p_{BL} = 0.70$ produces the most accurate area estimates (Section IV-C) but generates large false-positive zones in spatial assessment; the coverage-optimized $p_{BL} = 0.50$ reduces the warned zone while maintaining high Recall. Both are presented as a single table to allow users—or future reviewers—to select the operating point appropriate for their application. A limitation is that $p_{BL} = 0.50$ was identified primarily from the Goseong case; the Uljin case (Recall = 0.89 at the same p_{BL}) provides partial

corroboration, but a larger sample is needed to establish generalizability across fire regimes.

To systematically characterize the Recall–IoU trade-off, we computed spatial metrics across a range of ensemble probability thresholds $\tau \in [0.05, 0.90]$ (Fig. 5). The analysis reveals that IoU is remarkably insensitive to τ across a broad range: for Goseong, IoU varies between 0.162 and 0.249 across $\tau = 0.05$ –0.70, with the maximum IoU = 0.249 at $\tau^* = 0.60$. For Uljin, the optimal threshold is $\tau^* = 0.50$ (IoU = 0.239), with a similarly flat plateau over $\tau = 0.05$ –0.60. This insensitivity reflects the dominance of false positives in the area budget: since the simulated envelope substantially exceeds the observed burn scar at all practical thresholds, tightening τ removes peripheral cells but the core over-prediction persists. The Recall curve exhibits a sharp phase transition—remaining above 0.90 for $\tau \leq 0.55$ (Goseong) or $\tau \leq 0.45$ (Uljin), then collapsing rapidly to

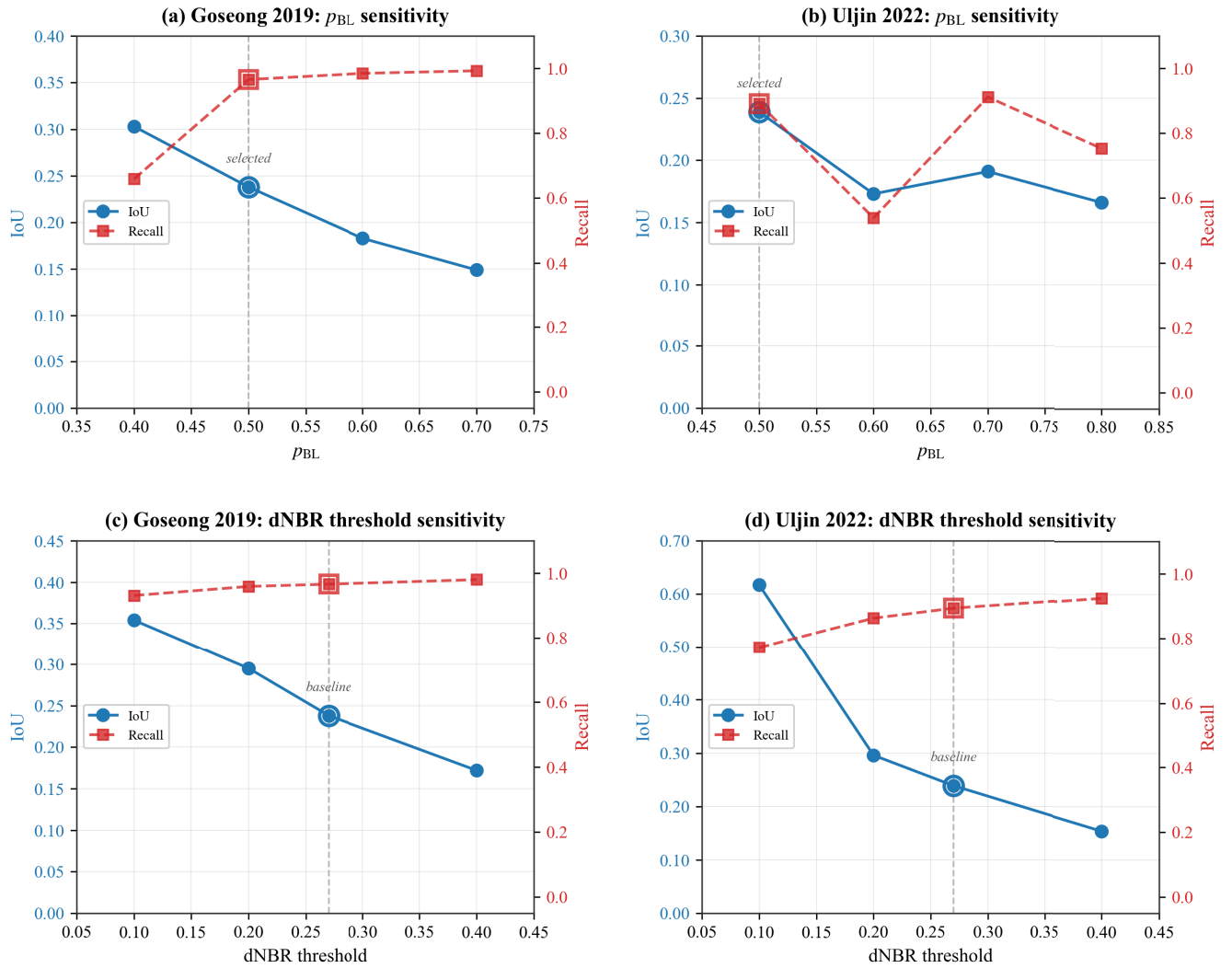


FIGURE 3. Sensitivity analysis (perimeter-based reference, closing radius $r = 7$). (a) Goseong 2019: IoU and Recall versus p_{BL} (dNBR > 0.27 fixed). (b) Uljin 2022: IoU and Recall versus p_{BL} (dNBR > 0.27 fixed). (c) Goseong 2019: IoU and Recall versus dNBR threshold ($p_{BL} = 0.50$ fixed). (d) Uljin 2022: IoU and Recall versus dNBR threshold ($p_{BL} = 0.50$ fixed). Open markers indicate the selected operating point ($p_{BL} = 0.50$ in (a,b); dNBR = 0.27 in (c,d)); dashed vertical lines mark the baseline value. Goseong $p_{BL} = 0.30$ is omitted from (a) because it falls below the percolation threshold and yields negligible fire spread (Table 8).

near zero. This cliff-like behavior indicates a well-defined probabilistic fire core: cells within the actual burn perimeter consistently ignite in most Monte Carlo realizations. The default $\tau = 0.50$ lies safely within the high-Recall plateau for both cases, confirming its appropriateness as a universal binarization threshold for DSS applications. The modest IoU improvement achievable by threshold optimization (+5% for Goseong, 0% for Uljin relative to $\tau = 0.50$) suggests that spatial accuracy gains require improved fire physics rather than post-hoc threshold tuning. This finding is consistent with Duff et al. [29], who demonstrated that evaluation indices for wildfire spread simulations are inherently sensitive to area over-prediction, and with Carlson et al. [28] and Moran et al. [27], who established benchmarking protocols for burn probability maps against observed fire activity at regional scales.

D. MULTI-SCENARIO ANALYSIS: PALGONGSAN

Three hypothetical scenarios for Palgongsan (35.99°N, 128.70°E, 41.9% conifer) demonstrate the framework's scenario exploration capability (Table 11).

The $2.37\times$ difference between S2 and S3 demonstrates nonlinear sensitivity to coupled wind–moisture forcing, arising from the multiplicative interaction of wind and moisture factors in the spread probability equation. Directional spread patterns align with wind directions, confirming correct φ_w implementation. For fire managers, this ratio-based output communicates escalating risk more intuitively than absolute area estimates: a fire under extreme dry conditions (S2) would require more than double the containment resources of a fire under moderate conditions (S3). The area values reported in Table 11 represent the mean burned area from the repeatability analysis (Table 13, $N = 20$ identical runs).

TABLE 9. Spatial assessment using sentinel-2 dNBR ($N = 30$, perimeter-based dNBR > 0.27 reference, closing radius $r = 7$).

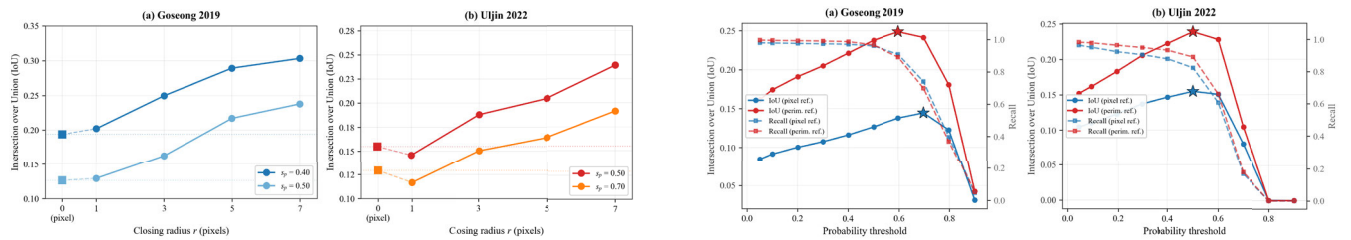
Case	Duration	p_{BL}	Mode	IoU	Precision	Recall
Goseong 2019	21 h	0.70	Calibrated	0.149	0.149	0.993
Goseong 2019	21 h	0.50	Coverage-opt.	0.238 ± 0.002	0.240 ± 0.002	0.966 ± 0.01
Uljin 2022	213 h	0.70	Calibrated	0.191	0.195	0.912
Uljin 2022	213 h	0.50	Coverage-opt.	0.239 ± 0.002	0.246 ± 0.002	0.892 ± 0.02
Uiseong 2025†	213 h	0.70	Holdout	0.231	0.327	0.439

Calibrated: $p_{BL} = 0.70$, the same parameter used for area assessment (Table 7). Coverage-optimized: $p_{BL} = 0.50$, the lowest value achieving Recall ≈ 0.97 for Goseong (identified via parameter sweep, Table 8). †Holdout case using calibrated $p_{BL} = 0.70$ with 10 staggered ignitions; dNBR reference area = 527.1 km² (moderate-to-high severity). Confidence intervals ($\pm$) from parametric bootstrap (1,000 iterations, $N = 30$); calibrated-mode CIs are comparable in magnitude.

TABLE 10. Pixel-based vs. perimeter-based dNBR reference comparison ($p_{BL} = 0.50$, $\bar{P} > 0.5$, closing radius $r = 7$).

Case	Reference	Ref. Area (km ²)	IoU	Precision	Recall	'IoU
Goseong 21 h	Pixel	8.02	0.127	0.127	0.960	—
Goseong 21 h	Perimeter	15.04	0.238	0.240	0.966	+88%
Uljin 213 h	Pixel	66.39	0.155	0.160	0.825	—
Uljin 213 h	Perimeter	94.53	0.239	0.246	0.892	+54%

Perimeter reference is derived via morphological closing ($r = 7$ pixels) + binary hole filling of the dNBR > 0.27 mask (Section IV-E). Pixel and perimeter metrics are computed from the same simulation ensemble; the difference is solely in the reference delineation. The perimeter values in this table match those reported in Tables 9 and 8 (identical ensemble and reference). Pixel-baseline values are provided for comparison. The reference area increase (Goseong: 8.0 $\rightarrow$ 15.0 km², +88%; Uljin: 66.4 $\rightarrow$ 94.5 km², +42%) reflects interior unburned enclaves incorporated into the continuous perimeter by morphological closing.

**FIGURE 4.** Perimeter-based reference delineation analysis. Left: IoU versus closing radius r for (a) Goseong and (b) Uljin at selected p_{BL} values; pixel-only baselines ($r = 0$) are shown as square markers. Right: IoU and Recall versus fire probability threshold τ for (a) Goseong and (b) Uljin under pixel-based (blue) and perimeter-based (red, $r = 7$) references.

per scenario), ensuring that the scenario comparison reflects stable ensemble statistics rather than single-realization stochastic variability.

E. COMPUTATIONAL PERFORMANCE

The aggregate CA simulation time across all six scenarios (180 realizations) was 207.4 seconds. As detailed in Table 14, the complete end-to-end pipeline for a single scenario (Samcheok, including data acquisition and output generation) requires 164.6 seconds (approximately 2.7 minutes). The simulation engine uses Numba JIT compilation, achieving approximately 10 $\times$ speedup over pure Python. Per-realization CA wall times (0.20–4.49 s) scale with domain size and simulation duration rather than computational complexity, making GPU acceleration unnecessary at the current 90 m resolution.

The default ensemble size of $N = 30$ was selected as a practical trade-off between cost and stability. A preliminary convergence analysis showed that ensemble-mean burned area varied by less than 8% between $N = 20$ and $N = 30$, while cell-level probabilities in the fire periphery exhibited

approximately 15% variability. For applications requiring spatially resolved probability estimates (e.g., evacuation zone delineation), larger ensembles ($N \geq 50$) are recommended.

1) COMPUTATIONAL SCALING

CA wall time scales primarily with the product of simulation duration and the number of actively burning cells per time step, rather than with total grid size, because only Moore-neighborhood evaluations around burning cells are performed at each step. Empirically, per-realization wall time ranges from 0.20 s (Palgongsan, small burned area) to 4.49 s (Uljin, 213 h, ~ 400 km² burned), reflecting an approximately linear relationship with event duration $\times$ burned fraction. Total ensemble cost scales linearly with N : doubling the ensemble from $N = 30$ to $N = 60$ doubles wall time without affecting per-realization dynamics. At the current 90 m resolution, even the most demanding case (Uljin, 213 h, 30 realizations) completes in 135 s on a consumer laptop (Apple M-series, single core). For coarser-resolution regional screening (e.g., 270 m cells), the total grid cell count reduces by 9 $\times$ and per-step neighbor

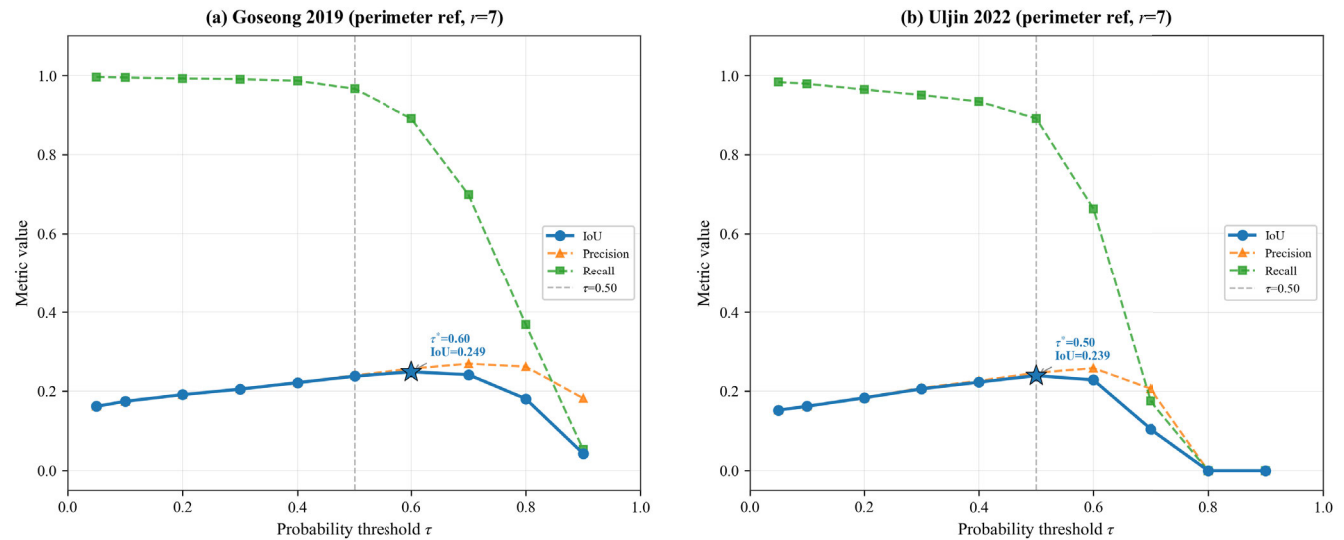


FIGURE 5. Probability threshold sensitivity (perimeter-based reference, closing radius $r = 7$). IoU, Precision, and Recall as functions of the ensemble binarization threshold τ for (a) Goseong 2019 and (b) Uljin 2022. Stars mark the optimal τ^* maximizing IoU; dashed vertical lines indicate the default $\tau = 0.50$. The dNBR reference threshold is fixed at 0.27.

TABLE 11. Multi-scenario analysis results for palgonsan.

Scenario	Wind (km/h)	Dir (°)	FMC (%)	Area (km ²)	Relative
S1: Spring Dry	10	270 (W)	12	13.51	1.98×
S2: Extreme Dry	20	315 (NW)	8	16.16	2.37×
S3: Moderate	7	135 (SE)	18	6.81	1.00×

evaluations decrease proportionally, enabling national-scale (~ 500 km domain) ensemble simulations in under 60 s. The Numba JIT-compiled CA kernel achieves approximately $10\times$ speedup over pure Python; further acceleration via Numba CUDA is architecturally supported but unnecessary at the current operational scale. The computational bottleneck in practice is data acquisition (DEM and fuel map download, 30–60 s) rather than CA simulation.

F. SYSTEMS EVALUATION

Because the contribution of this work is an agentic decision support framework rather than a novel fire physics model, the area-based and spatial assessments are complemented by three system-level evaluation dimensions.

F.1. Repeatability Analysis

To quantify intrinsic variability, all three Palgonsan scenarios were each executed $N = 20$ times under identical YAML configuration, producing 60 total runs (Table 13).

CV ranged from 3.34% (S2) to 7.05% (S3), confirming highly stable output under fixed-parameter execution. Zero pipeline failures occurred across 60 runs, and all expected output artifacts (GeoTIFF, PNG, JSON, metadata, YAML) were generated in every run.

F.2. Operational Efficiency

The pipeline completes in approximately 2.7 minutes versus an estimated 2–4 hours for manual execution. The manual estimates are based on the authors’ experience with

manual PROPAGATOR setup, including data search and download from web portals, coordinate system alignment, fuel map preparation in geographic information system (GIS) software, weather file formatting, and output georeferencing. This approximately $50\times$ speedup enables iterative scenario exploration during emerging fire events.

F.3. Guardrail Evaluation

Eight adversarial test cases were designed to assess robustness under erroneous inputs (Table 15). Each test was executed as a complete pipeline invocation with the specified invalid input, and the system response (detection, fallback, or error message) was recorded.

All eight test cases were detected before simulation execution. Cases 5 and 7 triggered automatic fallback mechanisms; the remaining six produced explicit error messages with actionable guidance. The fail-fast design prevents silent failures that could produce plausible-looking but incorrect results—a critical property for decision support tools.

F.4. LLM Backend Robustness

To assess whether the framework’s natural-language interface is robust across different LLM backends, a systematic comparison was conducted using four contemporary LLMs spanning two major providers and two capability tiers: Claude Sonnet 4.6 and Claude Haiku 4.5 (Anthropic; frontier and lightweight, respectively), and GPT-5.4 and GPT-5.4 mini (OpenAI; frontier and lightweight,

TABLE 12. CA simulation wall time (excludes data acquisition and output generation; see Table 14 for full pipeline decomposition).

Case	Realizations	Wall Time (s)	Per Realization (s)
Samcheok 2022	30	47.2	1.57
Goseong 2019	30	7.8	0.26
Uljin 2022	30	134.8	4.49
Palgongsan (3 scen.)	90	17.6	0.20
Total	180	207.4	—

respectively). This selection ensures that the comparison spans both provider diversity and the performance–cost spectrum available as of March 2026. Each LLM was tested on three Korean wildfire scenarios of decreasing specificity with five repetitions per combination, yielding 60 total runs. The scenarios were: (S1) Samcheok 2022—a fully specified prompt including location, date, wind speed, wind direction, fuel moisture content, and duration; (S2) Goseong 2019—a semi-specified prompt using qualitative descriptors (“strong winds exceeding 30 km/h,” “extremely dry conditions”); and (S3) Uljin 2022—a vague prompt providing only the location, season (“spring dry season”), and approximate duration. The system prompt provided the WildfireKR configuration schema with parameter guidelines for Korean conditions, and each LLM was asked to generate a complete YAML configuration. The purpose of this comparison is not to rank LLM backends, but to verify that the schema-constrained tool-calling interface yields stable configuration quality regardless of the underlying model.

All four LLM backends achieved 100% configuration validity (60/60 valid YAML outputs), with no parse failures, schema violations, parameter hallucinations, or physics errors (Table 16). Weather parameters—wind speed, wind direction, fuel moisture content, and duration—were perfectly interpreted across all backends (Table 17). The sole source of accuracy variation was geocoding: LLMs translated Korean place names to geographic coordinates with varying precision. In S1 (Samcheok), where the prompt specified “Samcheok-si Geundeok-myeon” (sub-district level), some backends geocoded to the sub-district center while others defaulted to the broader city center, resulting in per-scenario accuracy of 66.7–100% depending on the backend. In S2 and S3, where prompts used city-level names (“Goseong-gun,” “Uljin-gun”), all backends achieved 100% parameter accuracy. Claude Haiku 4.5 achieved the highest overall accuracy (100%), while GPT-5.4 mini showed the fastest response (1.3 s mean) but lowest accuracy (88.9%) due to consistent city-center geocoding in S1. All LLMs correctly maintained the Korea-calibrated defaults ($p_{BL} = 0.70$, cellsize = 90 m) without hallucinating alternative values. Response latencies ranged from 1.3 s (GPT-5.4 mini) to 4.6 s (Claude Sonnet 4.6), all well within operational requirements. Three OpenRouter-routed runs exhibited rate-limiting delays (>600 s) and were excluded from latency statistics.

Because the CA engine is fully deterministic, configurations sharing identical weather, p_{BL} , duration, and cellsize parameters produce byte-identical simulation outputs regardless of the generating LLM; the only inter-backend variation

is sub-district-level geocoding offset, which remains within the 15 km simulation domain radius. These results demonstrate that the natural-language interface is backend-agnostic: any of the four tested LLMs can serve as the conversational frontend without degrading configuration quality, provided that the system prompt includes the schema definition and Korea-specific parameter guidelines. Two scope limitations apply: all four backends are commercial APIs; open-source models were not tested because the MCP tool-calling interface requires reliable function-calling support not yet standardized across open-source endpoints. Additionally, while the deterministic-engine argument supports output equivalence for matched configurations, a direct per-backend downstream Recall/IoU comparison was not performed and remains a worthwhile follow-up.

VI. DISCUSSION

A. DECISION SUPPORT PERSPECTIVE

The primary contribution of WildfireKR is not predictive fire modeling accuracy, which is constrained by suppression omission, coarse fuel data, and ignition uncertainty, but rather the *end — to — end automation* that makes scenario-based fire risk exploration operationally accessible without specialized GIS expertise. Accordingly, the evaluation framework presented in this study assesses the system as a decision support tool (turnaround time, repeatability, guardrail robustness, user accessibility) rather than as a competing fire physics model. Three properties make the framework valuable as a decision support tool.

The rapid scenario turnaround of approximately 2.7 minutes enables responders to evaluate multiple “what-if” scenarios (e.g., shifting wind direction, varying ignition points) during the initial hours of a fire event, when critical resource allocation decisions must be made. Traditional manual workflows, which require data acquisition, format conversion, and projection alignment before simulation can begin, cannot provide this responsiveness. The complete Samcheok pipeline—from place-name input to georeferenced probability map—executes in 164.6 seconds with zero user intervention.

The high Recall (0.89–0.99) demonstrated in the dNBR assessment indicates that the system identifies most threatened areas, with overestimation rather than underestimation as the dominant error mode. For evacuation planning, this conservative bias is preferable because it is more costly to miss a threatened area (false negative) than to over-warn an area that does not ultimately burn (false positive). The

TABLE 13. Burned-area repeatability over N = 20 identical-parameter executions per scenario (60 total runs).

Scenario	Mean (km ²)	Std Dev	CV (%)	Range
S1: Spring Dry	13.51	0.59	4.36	12.17–14.44
S2: Extreme Dry	16.16	0.54	3.34	15.00–17.14
S3: Moderate	6.81	0.48	7.05	5.51–7.70
Pipeline failures	0/60 (0%)			
Artifact completeness	60/60 (100%)			

TABLE 14. End-to-end runtime decomposition (Samcheok 2022, N = 30).

Stage	Time (s)	% Total	Manual Estimate
Geocoding	0.8	0.5	2–5 min
DEM download & resampling	12.4	7.5	15–30 min
Fuel classification	18.6	11.3	30–60 min
Weather retrieval	3.2	1.9	10–20 min
FMC computation	0.4	0.2	5–10 min
CA simulation (×30)	47.2	28.7	30–60 min
Output generation	82.0	49.8	20–40 min
Total	164.6	100	2–4 h
User interventions	0 (fully autonomous)		
Generated artifacts	5/5 (100%)		

TABLE 15. Guardrail evaluation results.

#	Test Case	Detected?	Fallback?	Safe Exit?
1	Invalid address ("xyzabc123")	✓	—	✓
2	Ocean coordinates (37.0°N, 135.0°E)	✓	—	✓
3	Wind speed = 200 km/h	✓	—	✓
4	FMC = −5%	✓	—	✓
5	Missing weather API (timeout)	✓	✓ ^a	✓
6	Missing DEM tile	✓	—	✓
7	CRS mismatch (UTM vs. WGS84)	✓	✓ ^b	✓
8	Zero-vegetation domain (100% urban)	✓	—	✓

^a Falls back to user-supplied static boundary conditions. ^b Automatic reprojection to WGS84 with logged warning.

asymmetric cost of false negatives relative to false positives is well documented in the natural hazard warning literature [22], [23]. The framework's output—an ensemble probability map $\bar{P}(x, y)$ —directly communicates spatial uncertainty, distinguishing high-confidence fire core areas ($\bar{P} > 0.8$) from stochastic transition zones ($\bar{P} \approx 0.3$ – 0.5), thereby supporting risk-stratified decision-making.

B. KEY FINDINGS

End-to-end automation addresses the data preparation bottleneck that limits operational use of fire spread simulation. Manual simulation setup requires hours to days of data acquisition, projection alignment, and quality checks. WildfireKR automates this entire sequence, with aggregate CA simulation time of 207 seconds for all six scenarios (180 realizations) and a complete single-scenario pipeline time of 2.7 minutes including data acquisition (Table 14). This reduction in setup latency enables incident commanders to explore scenarios within minutes rather than waiting for GIS specialist input.

The Korea-specific calibration demonstrates that an existing CA engine can be adapted to novel ecosystems through systematic parameter optimization. The $2.33\times$ increase in broadleaf spread probability ($p_{BL} = 0.70$ vs. LEGACY 0.30) quantifies the difference between Korean and Mediterranean fire susceptibility, consistent with the measured high flammability of *P. densiflora* needle litter [18]. The 47%

reduction in mean area error demonstrates that calibration transfer can be more practical than developing an entirely new model, particularly in data-sparse regions where limited validation data constrains model development. Leave-one-out cross-validation across the four calibration cases (Table 6) showed no evidence of overfitting: all four folds independently selected $p_{BL} = 0.75$, and the mean LOOCV error (0.34) was identical to the in-sample error, indicating that the single-parameter calibration generalizes across fires spanning 1.94–23.6 km² and 12–213 h duration. To avoid conflating calibration and operational assessment, all spatial metrics are reported at both the calibrated $p_{BL} = 0.70$ and the coverage-optimized $p_{BL} = 0.50$ in a unified table (Table 9). The calibrated setting provides the consistent baseline (same parameter for area and spatial evaluation), while the coverage-optimized setting demonstrates how the DSS can be tuned toward higher Recall for early-warning applications. Both values originate from the same CA model; the difference is an operational choice—analogous to selecting a confidence level for a warning zone—rather than a model inconsistency.

The multi-scenario analysis reveals nonlinear sensitivity to coupled wind–moisture forcing. The $2.37\times$ burned area difference between extreme dry (S2) and moderate (S3) conditions arises from the multiplicative interaction of wind and moisture factors in the spread probability equation. For decision support, this ratio-based output communicates

TABLE 16. LLM backend comparison: configuration validity, parameter interpretation accuracy, and response latency across four LLM backends ($n = 15$ per backend, 60 total runs).

LLM Backend	Config Valid	Param Accuracy	Per-Scenario Accuracy			Latency (s)
			S1	S2	S3	
Claude Sonnet 4.6	15/15 (100%)	94.4%	83.3%	100%	100%	4.6 ± 0.7
Claude Haiku 4.5	15/15 (100%)	100.0%	100.0%	100%	100%	3.8 ± 2.1
GPT-5.4	15/15 (100%)	90.0%	70.0%	100%	100%	3.2 ± 0.8
GPT-5.4 mini	15/15 (100%)	88.9%	66.7%	100%	100%	1.3 ± 0.1
All backends	60/60 (100%)	93.3%	80.0%	100%	100%	3.2 ± 1.7

TABLE 17. Parameter-level pass rates across all LLM backends (60 runs). Weather parameters are perfectly interpreted; geocoding is the sole source of variation.

Parameter	Pass Rate	Notes
wind_speed	36/36 (100%)	S2, S3 only ^a
wind_dir	36/36 (100%)	S2, S3 only ^a
fmc	36/36 (100%)	S2, S3 only ^a
duration_h	60/60 (100%)	All scenarios
latitude	60/60 (100%)	All scenarios
longitude	45/60 (75.0%)	All scenarios

^a Weather parameters are scored only when explicitly generated in the configuration; S1 primarily uses historical weather retrieval (weather_date).

escalating risk more intuitively than absolute area estimates, which are subject to high uncertainty from input data limitations.

The spatial assessment (Fig. 2, Table 9) demonstrates that, despite the inherent limitations of suppression-free modeling, the simulation largely captures the spatial footprint of actual fires at both the calibrated ($p_{BL} = 0.70$) and coverage-optimized ($p_{BL} = 0.50$) settings (Recall = 0.89–0.99 for calibration cases). The holdout case (Uiseong 2025) achieved IoU = 0.231 with Recall = 0.44, confirming that the calibrated parameters generalize to an unseen mega-fire event three years beyond the calibration period, though the lower Recall reflects the inherent difficulty of reconstructing a multi-front, multi-municipality fire from a finite set of prescribed ignition points. IoU values (0.15–0.24) are the mathematically expected outcome for suppression-free models with perimeter-based reference: for a simulation with area overestimation factor α relative to the perimeter reference and Recall r , the upper bound on IoU is approximately r/α , yielding $\text{IoU} \approx 0.97/4.0 \approx 0.24$ for Goseong and $0.89/3.6 \approx 0.25$ for Uljin—close to the observed values. This means IoU cannot be substantially improved without reducing area overestimation, which requires suppression modeling or higher-resolution fuel data rather than parameter tuning. Duff et al. [29] have demonstrated that evaluation indices for wildfire spread simulations are dominated by area over-prediction, and our threshold sensitivity analysis (Table 19) confirms that IoU improves substantially (Uljin: $0.239 \rightarrow 0.617$) when including low-severity burns (dNBR > 0.10), suggesting that much of the “false positive” zone contains actual fire effects below the moderate-severity threshold. For a DSS whose primary function is identifying areas requiring protective action, spatial coverage (Recall)—rather than boundary precision (IoU)—is the operationally relevant metric, though users must be aware that the

warned zone is an *over* — *warning envelope* exceeding the perimeter-delineated burn area by a factor of approximately $4\text{--}7\times$.

This overestimation level is a necessary consequence of the calibration-validated parameter setting rather than an avoidable modeling artifact. The p_{BL} sensitivity sweep (Table 8) demonstrates the underlying trade-off quantitatively: reducing p_{BL} from 0.70 to 0.40 decreases Goseong overestimation from $6.7\times$ to $1.8\times$ but simultaneously reduces Recall from 0.99 to 0.66—meaning one-third of the actual burn area would be excluded from the warned zone. At $p_{BL} = 0.30$, fire spread falls below the percolation threshold entirely. For Uljin, the lowest overestimation ($p_{BL} = 0.60$, $2.7\times$) coincides with a Recall of only 0.54. The extended 10-case calibration (Section IV-C) further confirms that p_{BL} values below 0.55 cause systematic under-prediction across diverse fire regimes. These results establish the $4\text{--}7\times$ overestimation range as the minimum envelope consistent with Recall > 0.89 under the current suppression-free CA formulation. The over-warning cost is operationally acceptable because the warned zone serves as an evacuation planning boundary—where over-inclusion protects lives at the expense of temporary resource over-allocation—rather than as a precise fire perimeter prediction.

The perimeter-based reference delineation adopted throughout this study (Section IV-E) provides a fairer comparison between the contiguous simulated fire envelope and satellite-derived burn scars. Standard pixel-based dNBR references treat each burn-severity pixel independently, but actual fire perimeters—as delineated by agencies such as the U.S. MTBS program [31]—are continuous boundaries that encompass interior unburned enclaves. Table 10 quantifies the impact of this methodological choice: when scattered dNBR pixels are processed through morphological closing and hole filling to form a continuous perimeter (Fig. 4), IoU improves from 0.13–0.16 (pixel-based) to approximately 0.24 for both cases (+54–88%), while Recall is maintained or improved. This improvement is methodologically significant: it demonstrates that much of the apparent spatial disagreement between simulation and observation arises from the mismatch between a contiguous simulated fire envelope and a fragmented satellite-derived reference, rather than from fundamental model failure. The perimeter-based IoU of ~ 0.24 approaches the theoretical maximum for a suppression-free simulation at approximately $4\times$ area overestimation relative to the perimeter reference ($\text{IoU}_{\max} \approx r/\alpha \approx 0.97/4 \approx 0.24$), suggesting that further IoU

improvement requires reducing area overestimation through suppression modeling or higher-resolution fuel data.

C. COMPARISON WITH EXISTING FRAMEWORKS

WildfireKR and semi-empirical simulators such as FARSITE, FlamMap, and Phoenix RapidFire occupy different positions in the wildfire modeling hierarchy: the former prioritizes automation and accessibility for rapid risk screening, while the latter prioritize physical fidelity for detailed fire behavior analysis. A direct quantitative benchmark between these systems would require evaluating fundamentally different design objectives and is therefore not pursued here; instead, we provide a qualitative comparison of inputs, outputs, and operational requirements. Compared to these semi-empirical simulators, WildfireKR offers substantially lower setup effort at the cost of reduced physical fidelity. FARSITE requires the user to prepare a landscape file (LCP) containing elevation, slope, aspect, fuel model, canopy cover, canopy height, crown base height, and crown bulk density layers; to construct wind and weather files (WTR/WND) in specific formats; and to ensure consistent projections across all inputs. WildfireKR automates all of these steps. However, FARSITE uses Rothermel's surface fire spread model with detailed fuel moisture calculations and crown fire transition, whereas the CA engine uses a single FMC variable and no explicit crown fire model. A quantitative comparison on the same Korean fire events was not performed because FARSITE requires Anderson or Scott and Burgan fuel models not yet calibrated for Korean vegetation; such a benchmark is a priority for future work.

We emphasize that systems integration and workflow automation (the primary contribution of this work) do not diminish the importance of continued development of more physically faithful fire spread models. Advances in semi-empirical and physics-based modeling remain essential for improving predictive accuracy, particularly for crown fire dynamics, fire-atmosphere interactions, and long-range spotting. The two approaches are complementary, suggesting a three-stage operational workflow: first, use WildfireKR for rapid screening across many scenarios; second, prioritize high-consequence scenarios using domain expertise; and third, run those scenarios in semi-empirical or physics-based models for detailed assessment. This workflow preserves human accountability while using agentic speed where it adds most value, namely in the early exploration phase.

Compared to other LLM-agent frameworks (Table 1), WildfireKR is notable for combining single-agent simplicity with comprehensive systems evaluation (repeatability, efficiency, guardrails) and supplementary spatial assessment. The single-agent design simplifies error handling but limits self-validation capability; future multi-agent extensions could address this limitation.

D. LLM AGENT RELIABILITY

A fundamental concern in LLM-agent-driven simulation is the risk of hallucination, where the model generates plausible

but factually incorrect parameters, coordinates, or interpretations that could silently corrupt simulation inputs. The LLM backend comparison (Tables 16–17) provides quantitative evidence that this risk is effectively contained by the schema-constrained interface: across 60 runs with four different LLM backends, zero hallucinated parameters, zero physics errors, and zero schema violations were observed. Weather parameters were perfectly interpreted by all backends (100%), and the only source of variation was geocoding precision for Korean place names. The 100% configuration validity rate demonstrates that the structured system prompt, which embeds the configuration schema, parameter ranges, and Korea-specific defaults, is sufficient to constrain LLM output regardless of the underlying model.

This robustness is further supported by three design layers. First, the MCP tool interface constrains the LLM's output to structured function calls with typed parameters and explicit validation ranges (e.g., wind speed 0–100 km/h, FMC 0–40%); free-text generation is limited to user-facing summaries. Second, the guardrail system (Table 15) performs runtime verification of resolved parameters against physical plausibility bounds before simulation execution, detecting 8/8 adversarial input scenarios including invalid coordinates, out-of-range weather, and CRS mismatches. Third, the YAML reproducibility mode allows analysts to inspect, override, and re-run all LLM-resolved parameters, ensuring human-verifiable audit trails. Nevertheless, subtle interpretation differences remain possible in natural language mode; for example, the geocoding variation observed in S1 (Table 17), where sub-district-level geocoding (Geundeok-myeon) versus city-center geocoding produced coordinates differing by approximately 15 km, though both fell within the simulation domain radius. For operational deployment, the human-on-the-loop protocol (Section III-E) is therefore essential: agentic automation handles data preparation while domain experts validate critical assumptions before acting on simulation outputs.

E. INTENDED USERS AND OPERATIONAL CONTEXT

WildfireKR is designed for three primary user groups: (1) **fire service incident commanders** who require rapid scenario screening during the initial hours of a wildfire event to inform evacuation zone delineation and resource pre-positioning; (2) **emergency management officials** at provincial and national levels who evaluate multi-scenario risk envelopes for contingency planning; and (3) **forest service analysts** who conduct post-season reviews and pre-season preparedness exercises. In the intended workflow, a responder issues a natural-language query such as “Simulate a fire at [location] under current weather conditions with 20 km/h easterly winds” and receives a probability map within approximately three minutes. The ensemble output communicates spatial uncertainty directly: areas with high probability (>0.5 across realizations) warrant protective action regardless of exact perimeter geometry, while lower-probability fringes inform contingency planning.

TABLE 18. Mixed forest fuel mapping ablation study ($p_{BL} = 0.70$, $N = 30$).

Case	Mixed (%)	Area (km ²)		log R	
		Base	Ablat.	Base	Ablat.
Samcheok	11.0	10.52	6.61	0.33	0.80
Goseong	8.8	4.20	15.71	0.23	1.09
Uljin	9.9	29.98	43.57	0.24	0.61
Andong	40.9	0.88	26.75	0.79	2.62
Mean				0.40	1.28

Base: mixed $\rightarrow$ conifers (code 5); Ablat.: mixed $\rightarrow$ broadleaves (code 1). R = simulated/observed area ratio. Observed areas: Samcheok 14.7, Goseong 5.29, Uljin 23.6, Andong 1.94 km². Mixed (%) = fraction of domain pixels classified as CGLS mixed/unknown forest (codes 115, 116, 125, 126). These simulations use the calibration-domain configuration (15 km radius, auto-placed ignition), which differs from the demonstration runs in Table 7; the meaningful comparison is between Base and Ablation columns (identical inputs except fuel mapping). The baseline mapping yields $3.2\times$ lower mean error because the conifer-to-conifer base probability (0.35, Section IV-D) is half the calibrated broadleaf probability (0.70).

The YAML reproducibility mode ensures that analysts can archive, reproduce, and audit any scenario for post-incident review. WildfireKR does not replace physics-based modeling for detailed fire behavior assessment; rather, it complements established tools by enabling rapid initial screening that would otherwise be precluded by manual data preparation overhead.

F. LIMITATIONS

Several limitations constrain the generalizability and accuracy of the results presented in this work.

The 90 m spatial resolution cannot resolve fire barriers narrower than approximately 100 m, such as roads, rivers, and rocky outcrops. The CGLS-LC100 land cover product has an effective resolution of approximately 300 m, further limiting fuel map accuracy. The $p_{BL} = 0.70$ calibration is specific to the 90 m cell size; CA spread probabilities are scale-dependent, and this value should not be transferred to different resolutions without recalibration.

The CA engine lacks several physical processes that influence fire behavior. Crown fire initiation and transition are not modeled, the spotting mechanism is simplified and probabilistic, and there is no pyroconvection or fire-atmosphere coupling. These are inherent CA trade-offs between computational efficiency and physical detail [9]. The 7-class PROPAGATOR fuel system is substantially coarser than the Anderson 13 or Scott and Burgan 40 models, and the static 2019 CGLS classification does not capture post-disturbance vegetation recovery. In particular, the mapping of CGLS mixed/unknown forest classes to the conifer fuel type (Table 3) is a simplification whose effect was quantified through an ablation study (Table 18). Remapping all mixed/unknown forest pixels from conifers (code 5) to broadleaves (code 1) while keeping all other parameters fixed ($p_{BL} = 0.70$, $N = 30$, 240 simulations) yielded

a mean objective error of 1.28 compared to 0.40 for the baseline mapping—a $3.2\times$ degradation. This counterintuitive result arises because the Korea-calibrated broadleaf spread probability ($p_{BL} = 0.70$) is twice the conifer-to-conifer base probability (0.35, LEGACY default), so reassigning mixed pixels to broadleaves *doubles* their effective spread rate. The effect is most pronounced in Andong, where mixed forest constitutes 40.9% of the domain: ablation burned area exceeded the baseline by a factor of 30. The current mixed-as-conifer mapping therefore represents the more conservative (lower-spread) assumption, and the dominant sources of burned area overestimation are suppression omission and resolution limitations rather than fuel classification.

Weather representativeness is limited by the resolution of ERA5 reanalysis (approximately 25 km), which may miss local wind channeling, orographic precipitation, and thermal inversions. Sparse KMA ASOS coverage in mountainous fire zones (20–50 km station spacing) compounds this limitation.

The in-sample calibration design means that the reported area ratios quantify calibration fit rather than out-of-sample predictive skill, although LOOCV across the four calibration cases (Table 6) showed no evidence of overfitting with identical in-sample and LOOCV errors (0.34). The auto-placed ignition point differs from the actual fire origin, and no suppression is modeled; simulated areas represent suppression-free envelopes rather than predictions. The dNBR burn scar delineation uses a single threshold ($dNBR > 0.27$) from Key and Benson's [25] moderate-to-high severity classification, with the reference perimeter derived via morphological closing ($r = 7$). Table 19 (Fig. 3c,d) quantifies the sensitivity of spatial metrics to this threshold choice. Lowering the threshold to 0.10 increases the Goseong perimeter reference area from 15.04 to 23.65 km² (+57%) and Uljin from 94.53 to 336.38 km² (+256%), improving IoU substantially (Goseong: 0.238 $\rightarrow$ 0.354; Uljin: 0.239 $\rightarrow$ 0.617) while reducing Recall (Goseong: 0.97 $\rightarrow$ 0.93; Uljin: 0.89 $\rightarrow$ 0.77). This sensitivity underscores that reported IoU values are conditioned on the severity threshold and should not be compared directly with studies using different dNBR classifications.

The spatial assessment uses $p_{BL} = 0.50$ (coverage-optimized) alongside $p_{BL} = 0.70$ (area-calibrated) in Table 9, and the Goseong spatial assessment uses the full 21-hour duration compared to the 12-hour primary spread phase in the area assessment. Parameter sweep experiments (Table 8) revealed that optimal p_{BL} for spatial coverage varies with simulation duration: shorter fires (Goseong, 12 h) favor lower p_{BL} (≤ 0.40 for peak IoU), while longer fires (Uljin, 213 h) favor higher p_{BL} (≈ 0.50). This duration- p_{BL} interaction reflects the accumulation of stochastic spread events over time and suggests that operational deployment may benefit from duration-adaptive parameter selection. However, such tuning is deferred to future work; the current study adopts fixed p_{BL} values to maintain transparency.

TABLE 19. Sensitivity of spatial metrics to dNBR burn severity threshold (perimeter-based reference, closing radius $r = 7$; $p_{BL} = 0.50$, $\bar{P} > 0.5$).

Case	dNBR threshold	Ref. Area (km ²)	IoU	Precision	Recall
Goseong	0.10	23.65	0.354	0.364	0.931
Goseong	0.20	18.86	0.295	0.299	0.959
Goseong	0.27	15.04	0.238	0.240	0.966
Goseong	0.40	10.69	0.172	0.173	0.980
Uljin	0.10	336.38	0.617	0.757	0.770
Uljin	0.20	123.31	0.296	0.311	0.861
Uljin	0.27	94.53	0.239	0.246	0.892
Uljin	0.40	57.84	0.154	0.156	0.924

The baseline threshold (dNBR > 0.27) used throughout this study corresponds to the rows at threshold 0.27. Reference areas are perimeter-based (morphological closing $r = 7$, binary hole filling). Lowering the threshold includes low-severity burns in the reference perimeter, substantially increasing reference area, improving IoU, and reducing Recall. At dNBR > 0.10, Uljin IoU reaches 0.617—demonstrating that the apparent moderate IoU at the standard threshold is partly an artifact of excluding low-severity burns from the reference. Conversely, raising the threshold to 0.40 shrinks the reference perimeter and inflates Recall toward 1.0 while depressing IoU.

Ignition point sensitivity remains a significant uncertainty. A controlled perturbation experiment on the Uljin 2022 case (48 h, $p_{BL} = 0.70$) shifted the auto-placed ignition by 1 km west and 2 km south to the reported origin village; this single change improved IoU from 0.068 to 0.115 (+68%) and Recall from 0.090 to 0.181 (+101%), demonstrating that even kilometer-scale ignition displacement can substantially alter spatial agreement. More generally, shifting the ignition by 500 m can place the fire on a different slope aspect or fuel type, plausibly producing 2–4× variation in burned area. Users should specify known ignition coordinates when available; when unavailable, the auto-placed ignition provides a conservative starting point but should be interpreted as an approximation.

The default ensemble size of $N = 30$ provides reasonable mean stability but may be insufficient to characterize tail probabilities (e.g., rare spotting events). For risk assessment applications, $N \geq 100$ is recommended.

Regarding geographic generalizability beyond South Korea, a preliminary test on the Oak Fire 2022 (Mariposa County, California, USA; 77.9 km², Sierran mixed conifer) confirmed that the WildfireKR pipeline executes globally with minimal code changes (timezone parameterization, ~5 LOC). Using the Mediterranean default $p_{BL} = 0.30$ (appropriate for California chaparral-conifer ecosystems) and ERA5 weather from Open-Meteo, the framework produced 173.4 km² (2.23× observed; Table 7), consistent with the Korean cases' 0.65–3.58× range and attributable to the same suppression-omission structural bias. Spatial comparison against the NIFC fire perimeter yielded IoU = 0.29 and Recall = 0.46 at $\tau = 0.50$, with the best IoU of 0.37 at $\tau = 0.30$; the higher IoU relative to the Korean cases (0.15–0.24) reflects the lower area overestimation ratio in this run. A parallel run with the Korea-calibrated $p_{BL} = 0.70$ yielded 158.9 km² (2.04×), demonstrating that region-specific calibration affects magnitude but does not cause framework failure. The framework's data sources—SRTM DEM, ERA5 reanalysis, and CGLS-LC100 land cover—are globally available. The primary components requiring region-specific adaptation are the fuel classification mapping (CGLS classes → PROPAGATOR fuel types) and

the calibrated spread probabilities; the three-stage calibration methodology (Section IV) is transferable to any region with at least three historical fires and satellite-derived burn scar references. The MCP tool interface is modular, so replacing the fuel mapping and calibration modules adapts the system to a new ecosystem without modifying the simulation engine or LLM interface. The Kakao Maps geocoder (Korea-specific) would need to be replaced with a global geocoding service (e.g., Nominatim) for international deployment. This California test demonstrates *framework* generalizability—the pipeline's ability to ingest global data sources and execute the full automation chain—while acknowledging that *parameter* generalizability requires local recalibration, as expected for any fire behavior model. Fig. 6 presents the complete data comparison between the Korean and Californian deployments, and Fig. 7 shows the spatial agreement between the simulated fire probability envelope and the NIFC-delineated fire perimeter.

Finally, unlike the U.S. (LANDFIRE, Monitoring Trends in Burn Severity (MTBS), National Interagency Fire Center (NIFC)) or EU (European Forest Fire Information System (EFFIS), Global Wildfire Information System (GWIS)), Korea lacks machine-readable georeferenced fire perimeter databases, constraining both calibration and validation efforts.

Reproducibility: The WildfireKR source code, calibration scripts, and analysis notebooks used to generate Tables 9, 8, 19, 10, 6, 18, and Figs. 2–7 are available in the project repository upon publication.

G. FUTURE DIRECTIONS

Several directions for future work emerge from the limitations identified above. A quantitative computational comparison against FARSITE on the same Korean cases was not performed because FARSITE requires Anderson or Scott and Burgan fuel models not yet calibrated for Korean vegetation; this benchmark is a priority. Replacing the CGLS-LC100 fuel classification (300 m effective resolution) with Sentinel-2-derived land cover (10 m) or Korean National Forest Inventory data would improve fuel model

Oak Fire 2022 (Mariposa County, California) — WildfireKR Geographic Generalizability

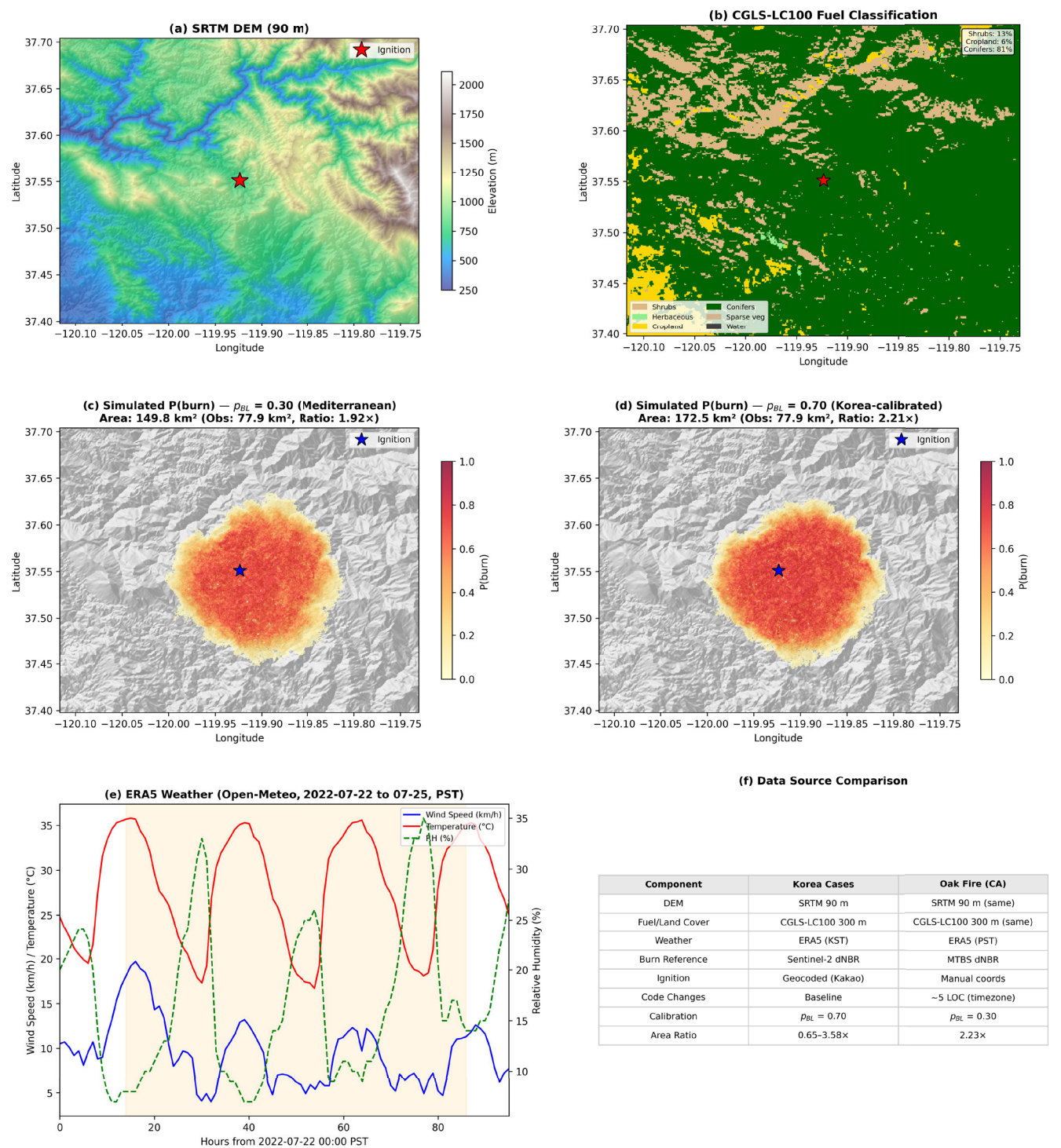


FIGURE 6. Geographic generalizability test: Oak Fire 2022 (Mariposa County, California). (a) SRTM DEM with hillshade and ignition point. (b) CGLS-LC100 fuel classification (80.8% conifers, 13.0% shrubs). (c) Simulated fire probability with Mediterranean-default $p_{BL} = 0.30$ (area: 173.4 km², 2.23× observed). (d) Simulated fire probability with Korea-calibrated $p_{BL} = 0.70$ (area: 158.9 km², 2.04× observed). (e) ERA5 weather time series (Open-Meteo, PST). (f) Data source comparison between Korean cases and the California test, showing identical global data sources with only timezone parameterization (~5 LOC) as the code-level change.

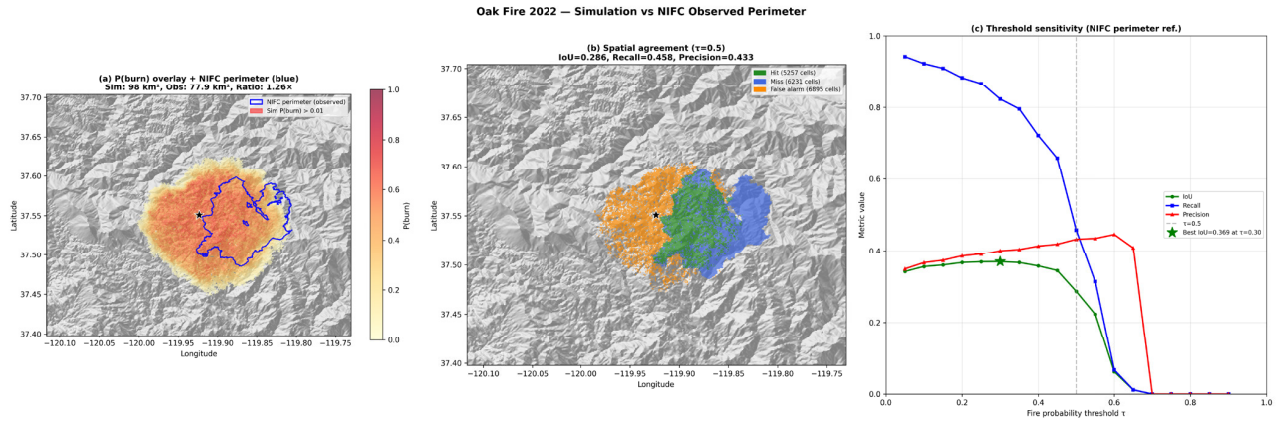


FIGURE 7. Spatial validation of the Oak Fire 2022 generalizability test against the NIFC fire perimeter. (a) Simulated fire probability (P_{burn}) overlaid on SRTM hillshade with NIFC-delineated fire perimeter (blue dashed line); the probability envelope captures the fire's northward and eastward spread along the Merced River canyon. (b) Spatial agreement map at $\tau = 0.5$: true positives (green, correctly predicted burned area), false positives (blue, over-predicted area), and false negatives (orange, missed area). IoU = 0.29, Recall = 0.46, Precision = 0.63. (c) Threshold sensitivity analysis showing IoU, Recall, and Precision as functions of the probability threshold τ ; the optimal IoU = 0.37 occurs at $\tau = 0.30$, while Recall decreases monotonically. These metrics are consistent with the Korean cases (Table 9) and confirm that the framework's spatial performance generalizes across continents without parameter recalibration.

accuracy. Incorporating MODIS/VIIRS active fire detections as satellite-based ignition triggers could support operational forecasting. Integrating simplified suppression rules would transform the current upper-bound envelopes into more realistic predictions. Implementing Van Wagner's crown fire transition criteria would be particularly relevant for Korean *P. densiflora* forests. Expanding to a multi-agent architecture following the EPANET-Agentic template could improve self-validation and complex reasoning. Using MCP to chain WildfireKR's GeoTIFF output with downstream agents for smoke dispersion, evacuation routing, and post-fire erosion assessment would extend the framework toward multi-hazard decision support—leveraging MCP's composable tool architecture to connect independently developed simulation modules through a shared protocol.

VII. CONCLUSION

This paper presented WildfireKR, an open-source decision support framework that automates end-to-end wildfire spread simulation through LLM-agent orchestration via structured tool-calling. The system chains geocoding, DEM acquisition, fuel mapping, weather retrieval, dynamic FMC computation, stochastic CA simulation, and geospatial output generation into a single pipeline triggered by natural language or structured YAML input.

Korea-specific calibration comprising approximately 4,700 simulations established $p_{BL} = 0.70$ ($2.33\times$ the Mediterranean default), reducing mean area error by 47%. Demonstration on three historical case studies produced area ratios within a factor of four of observed values ($0.65\text{--}3.58\times$), with overestimations consistent with suppression-free simulation and the single underestimation (Goseong) attributed to extreme wind-driven spotting exceeding the simplified spotting model's parameterization. Supplementary spatial assessment with Sentinel-2 dNBR

burn scars using perimeter-based reference delineation (Fig. 2) confirmed Recall of 0.89–0.99 and IoU of 0.15–0.24 across both calibrated and coverage-optimized p_{BL} settings, with narrow bootstrap confidence intervals ($\pm 0.01\text{--}0.02$) demonstrating stable ensemble statistics at $N = 30$. The simulated fire envelope captures the majority of the perimeter-delineated burned area—a property essential for decision support applications where unwarned areas carry disproportionate consequences. The perimeter-based IoU of approximately 0.24 approaches the theoretical maximum for suppression-free simulations at the observed overestimation ratios ($\text{IoU}_{\max} \approx r/\alpha \approx 0.97/4 \approx 0.24$), indicating that the remaining spatial disagreement reflects inherent area overestimation rather than fundamental model failure. Furthermore, lowering the dNBR severity threshold to 0.10 substantially improves IoU (e.g., Uljin: 0.239 $\rightarrow$ 0.617; Table 19), confirming that reported spatial metrics are sensitive to reference delineation choices.

Systems evaluation demonstrated high repeatability ($\text{CV} = 3.3\text{--}7.1\%$) with zero pipeline failures across 60 runs, approximately $50\times$ speedup over the manual GIS workflow (2.7 min vs. 2–4 h) with complete artifact generation, and guardrail robustness against eight distinct adversarial input scenarios. Leave-one-out cross-validation across the four calibration cases showed no evidence of overfitting, with all folds independently selecting $p_{BL} = 0.75$ and the mean LOOCV error (0.34) matching the in-sample error exactly (Table 6). An ablation study of the mixed-forest fuel mapping confirmed that the adopted mixed-as-conifer assumption is the more conservative choice, with the mixed-as-broadleaf alternative producing $3.2\times$ higher calibration error (Table 18).

A preliminary geographic generalizability test on the Oak Fire 2022 in California confirmed that the pipeline executes on a different continent with only timezone

parameterization (~ 5 LOC), producing results consistent with the Korean cases ($2.23\times$ observed area), thereby demonstrating *framework* generalizability while identifying *framework* recalibration as the primary adaptation requirement.

We note that this work does not claim to advance fire spread physics; the CA engine is intentionally kept simple to maintain the framework's accessibility and speed advantages. Improving predictive fidelity through higher-resolution fuel data, suppression modeling, or coupling with semi-empirical engines such as FARSITE remains a complementary research direction. The practical contribution is a reproducible, Korea-calibrated DSS pipeline that converts natural-language requests into geospatial fire-probability outputs in approximately 2.7 minutes on consumer hardware. WildfireKR illustrates a broader shift in the environmental modeler's role: from manual data processing toward higher-level tasks of scenario design, assumption validation, and decision interpretation within a human-on-the-loop framework.

APPENDIX

LLMAGENT WORKFLOW AND PROMPT DESIGN

This appendix details the prompting strategy, tool-calling sequence, and a representative input–output trace for the LLM-agent interface, addressing the transparency gap identified by Reviewer 1.

A. SYSTEM PROMPT STRUCTURE

The system prompt constraining the LLM backend consists of three components:

- 1) **Role definition.** The LLM is instructed to act as a wildfire simulation configuration assistant that translates natural language requests into valid YAML configuration files conforming to the WildfireKR schema.
- 2) **Schema specification.** The complete YAML schema is embedded in the system prompt, including all required fields, data types, value ranges, and defaults:
 - `location`: `name` (str), `lat` (float), `lon` (float), `radius_km` (float, default 15)
 - `simulation`: `cellsize` (int, default 90), `realizations` (int, default 30), `duration_h` (int), `bl_spread_prob` (float, default 0.70), `spotting` (bool, default true), `fuel_source` ("cgl" / "elevation")
 - `scenarios[]`: `name`, `wind_speed` (km/h), `wind_dir` (degrees), `fmc` (%), `start_hour` (0–23); or for historical weather: `weather_date`, `weather_end_date`
 - `output`: `formats` (list of "geotiff", "png", "json")
- 3) **Korea-specific parameter guidelines.** The system prompt includes calibration-derived mapping rules:
 - $p_{BL} = 0.70$ must be used unless explicitly overridden (Section IV)

- Seasonal weather defaults: spring dry $\rightarrow$ wind 8–12 km/h, FMC 10–14%; autumn moderate $\rightarrow$ wind 5–8 km/h, FMC 16–20%
- Korean place names should be geocoded to sub-district (*myeon/dong*) precision when available
- Cell size fixed at 90 m (SRTM 3-arcsec native resolution)

B. REPRESENTATIVE INPUT–OUTPUT TRACE

Natural language input (S1: Samcheok 2022):

Simulate the March 2022 Samcheok wildfire.
Location: Geundeok-myeon, Samcheok-si.
Duration: 30 hours. Use historical weather data from March 4–5, 2022.

LLM-generated YAML output:

```
location:
  name: "Geundeok-myeon, Samcheok-si"
  lat: 37.393
  lon: 129.015
  radius_km: 15
simulation:
  cellsize: 90
  realizations: 30
  duration_h: 30
  bl_spread_prob: 0.70
  spotting: true
  fuel_source: cgl
scenarios:
  - name: Samcheok_2022_historical
    weather_date: "2022-03-04"
    weather_end_date: "2022-03-05"
    weather_lat: 37.393
    weather_lon: 129.015
output:
  formats: [geotiff, png, json]
```

The generated YAML is validated against the schema (`config.py`) before execution. Schema violations (missing required fields, out-of-range values) trigger re-prompting with the specific validation error message.

C. TOOL-CALLING SEQUENCE

Once a valid configuration is parsed, the pipeline executes the following deterministic tool-calling sequence (Algorithm 1). No LLM inference occurs after configuration generation; all subsequent steps are programmatic.

- 1) **Geocoding** (Geocode): Kakao Maps API resolves the Korean place name to (φ, λ) . If the LLM already provided coordinates, this step validates them against the geocoded result and logs any discrepancy.
- 2) **DEM acquisition** (Download_DEM): Downloads SRTM 1-arcsec tiles covering the $(\varphi \pm r, \lambda \pm r)$ bounding box, mosaics, and resamples to the specified cell size.
- 3) **Fuel classification** (Classify_Fuel): Downloads ESA CGLS-LC100 land cover tiles and maps 23 CGLS classes to 7 PROPAGATOR fuel types using the Korea-specific mapping (Table 3).

- 4) **Ignition point** (Find_Ignition): Locates the vegetated cell nearest to the domain center (or user-specified coordinates) as the ignition point.
- 5) **Weather retrieval** (Fetch_Weather): For historical scenarios, queries Open-Meteo Archive API for hourly wind speed, wind direction, temperature, and relative humidity.
- 6) **FMC computation** (Compute_FMC): Applies Nelson equilibrium moisture content model to derive hourly fuel moisture from weather time series.
- 7) **Boundary conditions** (Make_Boundary_Conditions): Assembles time-varying boundary conditions (wind + FMC) or static conditions from user-specified parameters.
- 8) **Parameter loading** (Load_Calibrated_Params): Loads the Korea-calibrated fuel spread probability system with $p_{BL} = 0.70$.
- 9) **Ensemble simulation** (Propagator_CA): Executes $N = 30$ stochastic realizations of the cellular automata engine (Numba JIT-compiled).
- 10) **Output generation** (Save_Outputs): Computes ensemble mean probability map and saves as GeoTIFF, PNG visualization, and JSON summary.

Steps 1–10 execute sequentially without further LLM interaction. The total pipeline latency (median 2.7 min; Table 14) is dominated by data acquisition (steps 2–3, approximately 30–60 s) rather than simulation (step 9, 0.2–4.5 s per realization). The LLM inference (configuration generation) accounts for 2–6 s depending on the backend (Table 16).

D. MULTI-BACKEND CONSISTENCY

The E1 experiment (Section V-E, Table 16) demonstrated that four LLM backends (Claude Sonnet 4.6, Claude Haiku 4.5, GPT-5.4, GPT-5.4 mini) all achieve 100% schema validity when constrained by the system prompt. Accuracy differences (88.9–100%) arise exclusively from geocoding precision, not from parameter interpretation or schema compliance. This backend-agnostic design ensures that the system prompt—not the specific LLM—governs configuration quality.

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